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Miller et al.

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(54) **PILE CONNECTION APPARATUS AND METHOD FOR SUPPORTING A VERTICAL PILE**

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(73) Assignee: **Rapid Footings Group, LLC**, Youngsville, NC (US)

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E02D 5/54 (2006.01)
E02D 27/50 (2006.01)

(52) **U.S. Cl.**
CPC ***E02D 5/54*** (2013.01); ***E02D 27/50*** (2013.01); *E02D 2600/30* (2013.01)

(58) **Field of Classification Search**
CPC .. E02D 5/54; E02D 5/80; E02D 27/12; E02D 27/14
See application file for complete search history.

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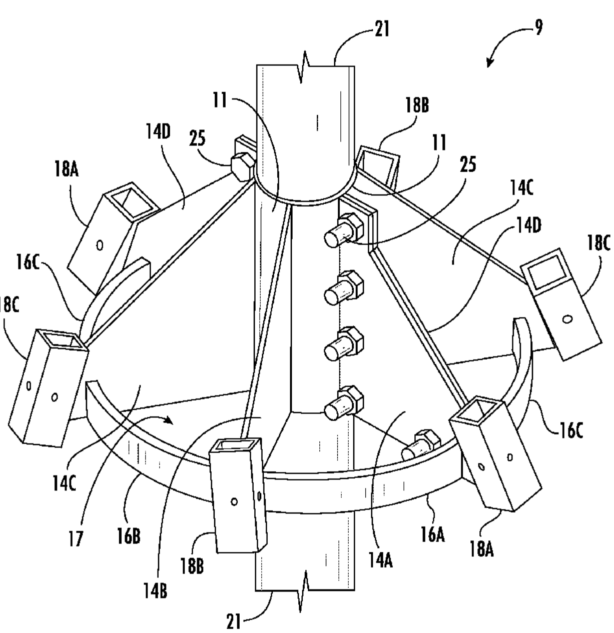
(57) **ABSTRACT**

A pile connection comprises at least one fixture having a collar portion, a plurality of micropile sleeves, a plurality of gusset members extending between the collar portion and the micropile sleeves, and at least one connecting member that couples adjacent gusset members and that is spaced apart from the collar portion. A recess bounded by a wall of the collar portion is configured to contact a portion of a vertical pile. One or more fasteners may be used to press the wall against the vertical pile, with certain embodiments employing two complementary fixtures that may be coupled to one another (e.g., using the fasteners) along opposing sides of the vertical pile. Each micropile sleeve is configured to receive a micropile that may be driven through the sleeve into the ground at a position spaced apart from the vertical pile.

22 Claims, 18 Drawing Sheets



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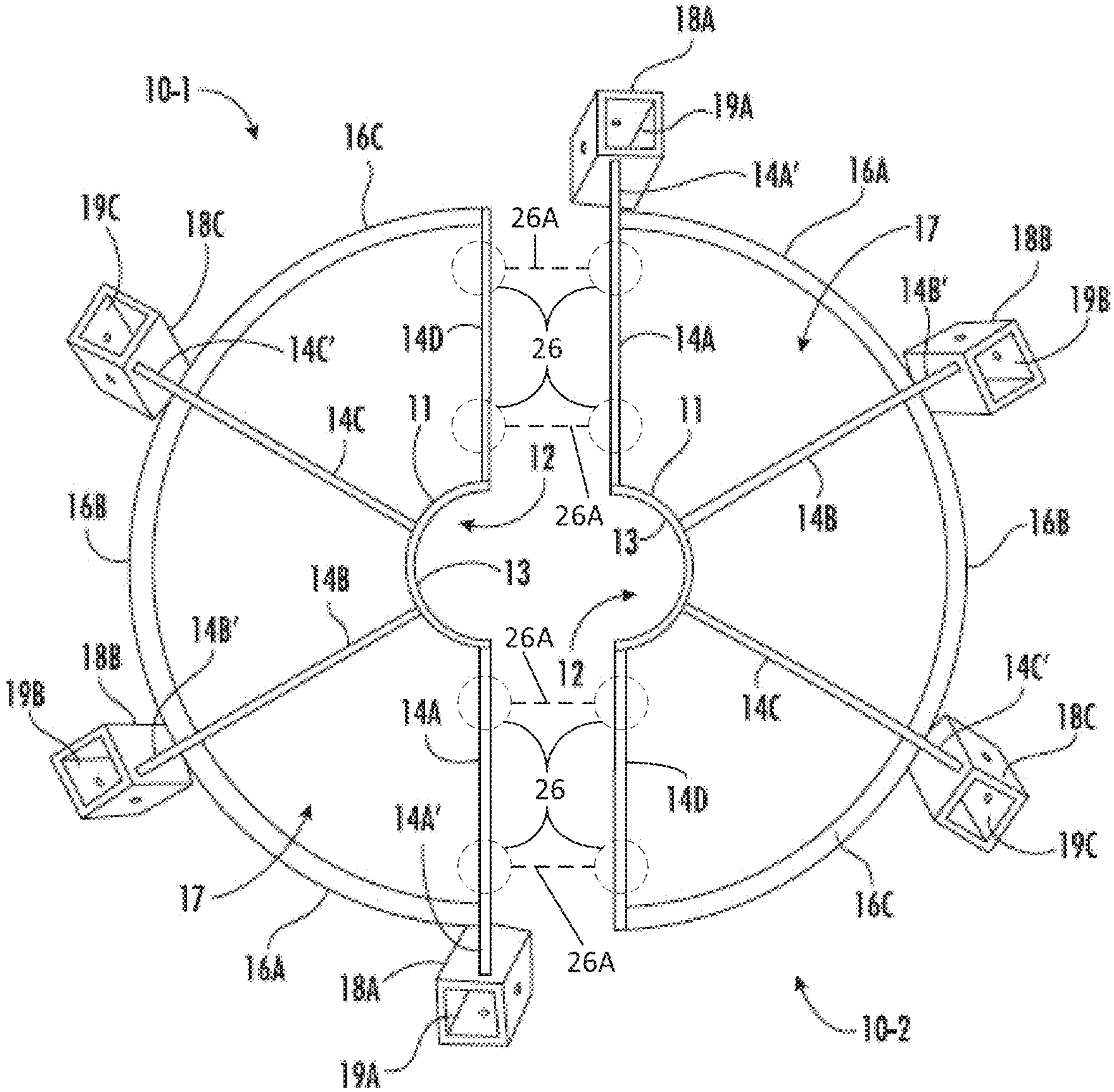
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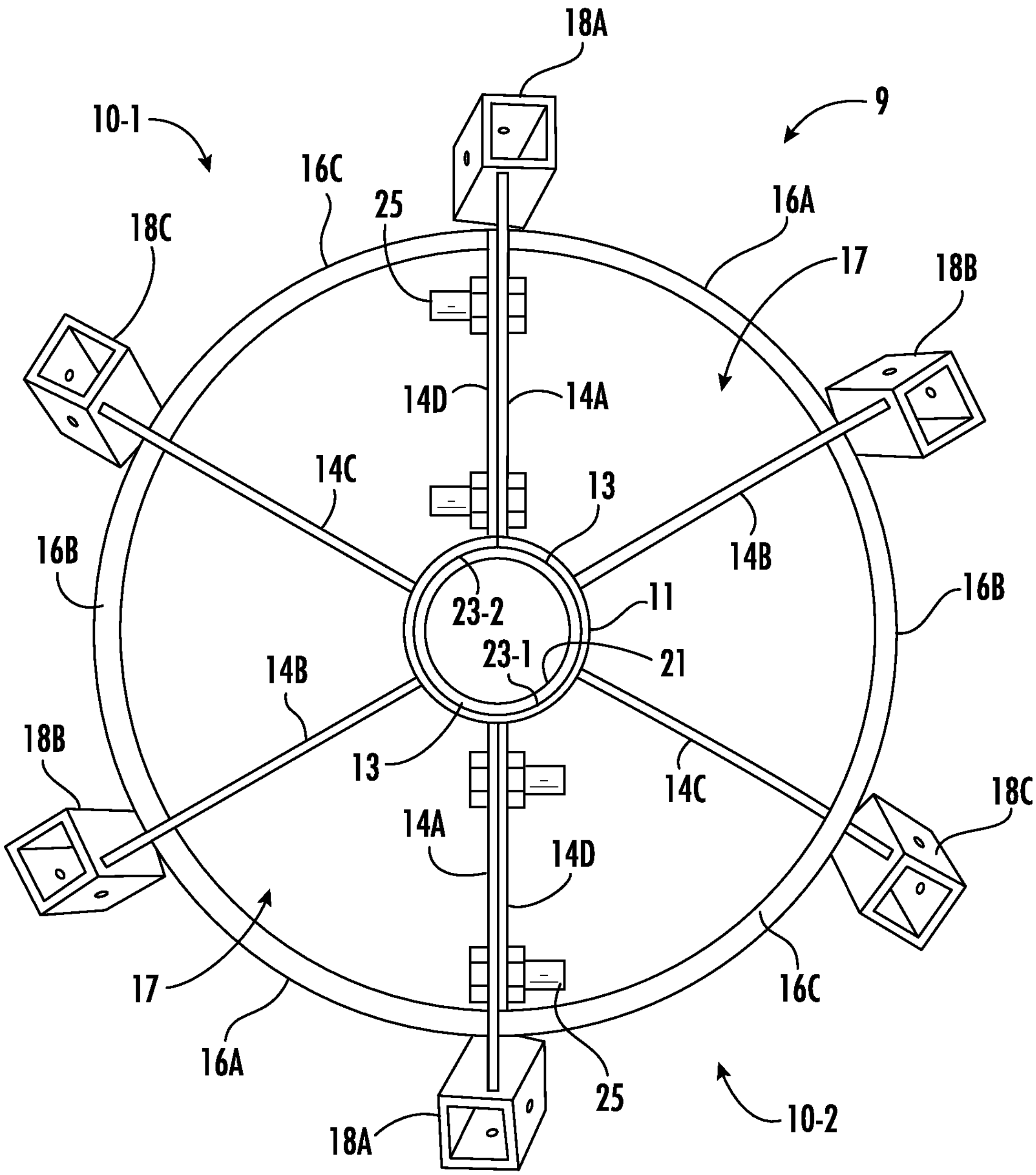
19A
14A'
16A
10-1
16C
26A
17
18B
19C
18C
14B'
19B
14D
26
14A
14C'
14C
11
11
12
16B
26A
14B
16B
13
13
14B
26A
14B'
12
14C'
18B
14A
14C
18C
19B
26
14D
14A'
19C
17
16C
16A
26A
18A
10-2
19A

FIG. 1A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9
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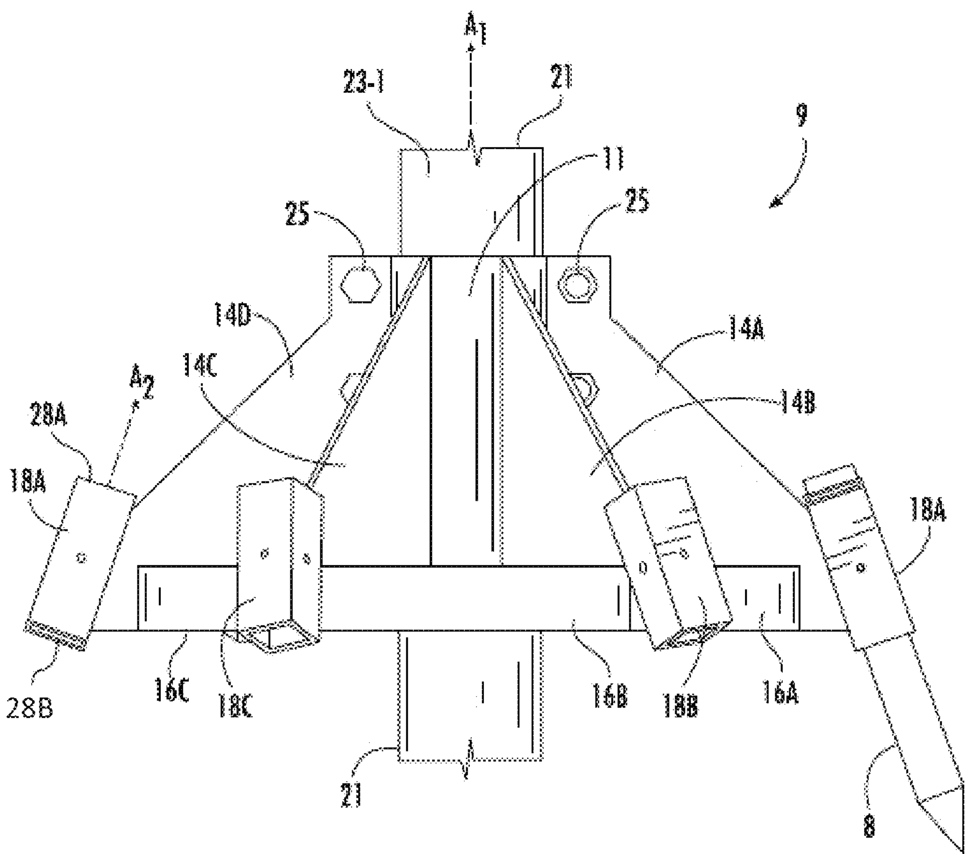
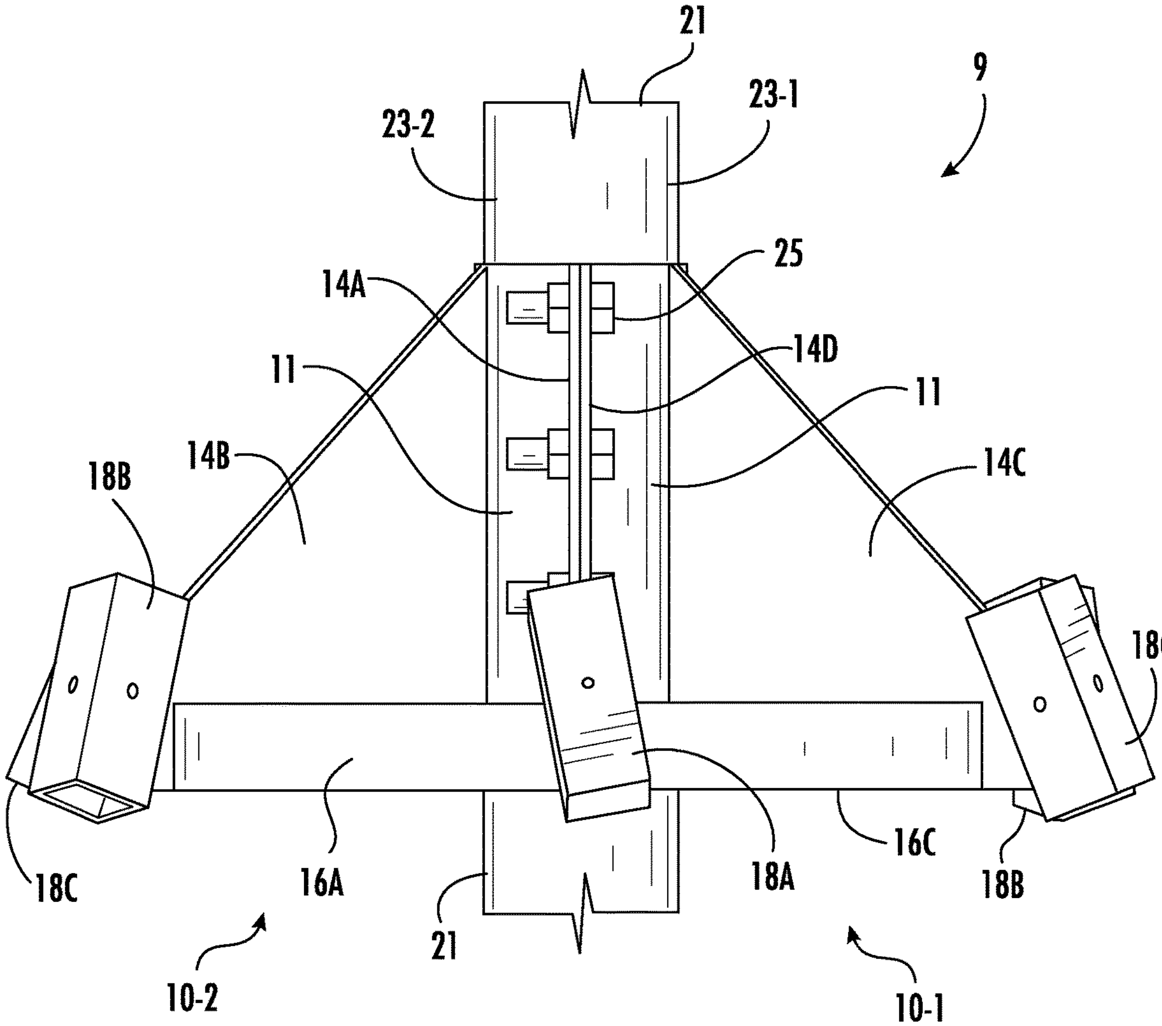
16C
25
16A
18C
17
18B
14D
14A
14C
13
16B
14B
16B
23-2
11
23-1
21
14B
18B
13
14C
18C
14A
14D
17
16C
16A
25
10-2
18A

FIG. 1B



A_1
23-1
21
9
11
25
25
14D
14A
A_2
14C
14B
28A
18A
18A
28B
16C
18C
16B
18B
16A
21
8

FIG. 1C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23-2
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14A
14D
11
11
14B
14C
18B
18C
18C
18A
16C
16A
18B
21
10-2
10-1

FIG. 1D

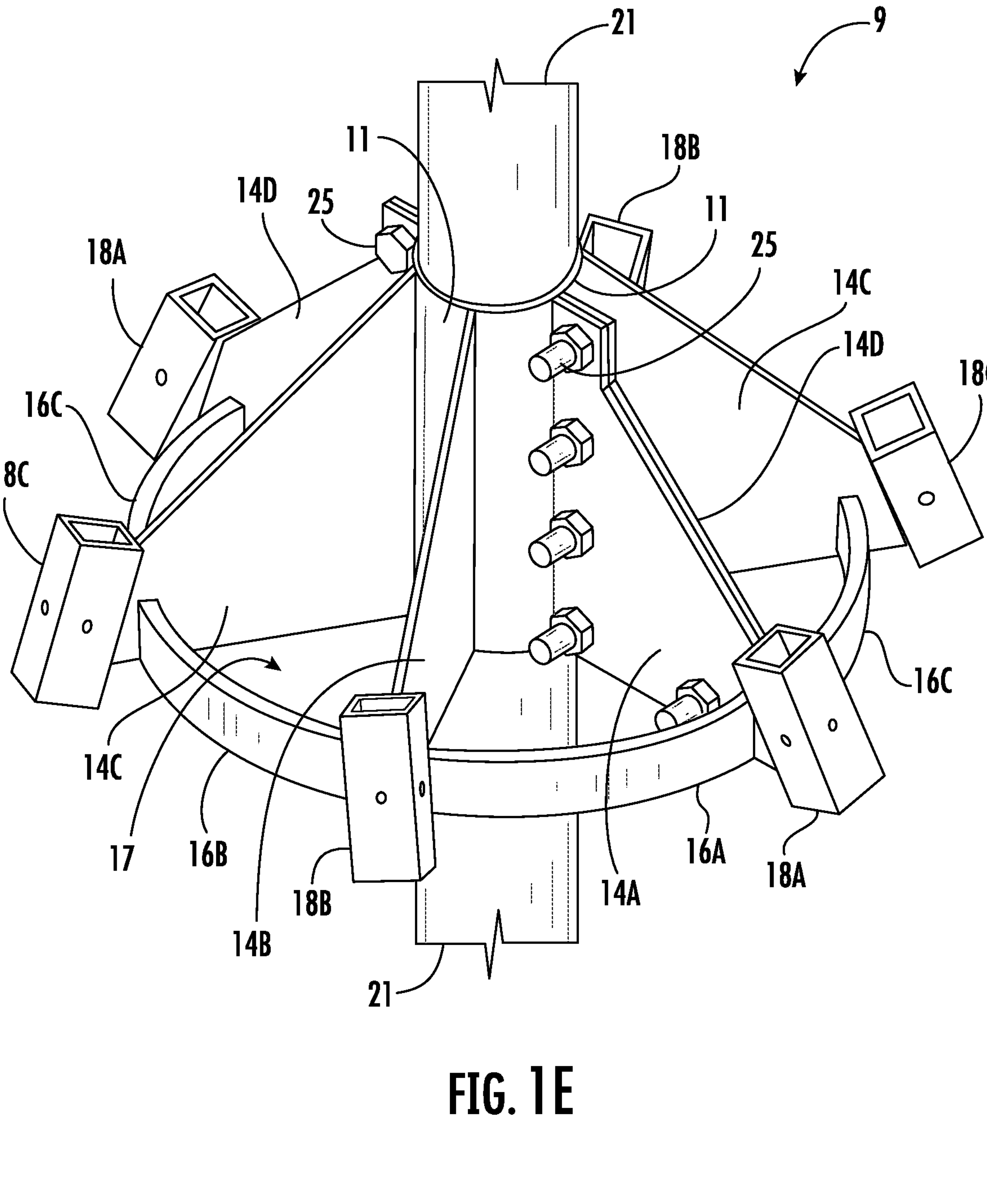
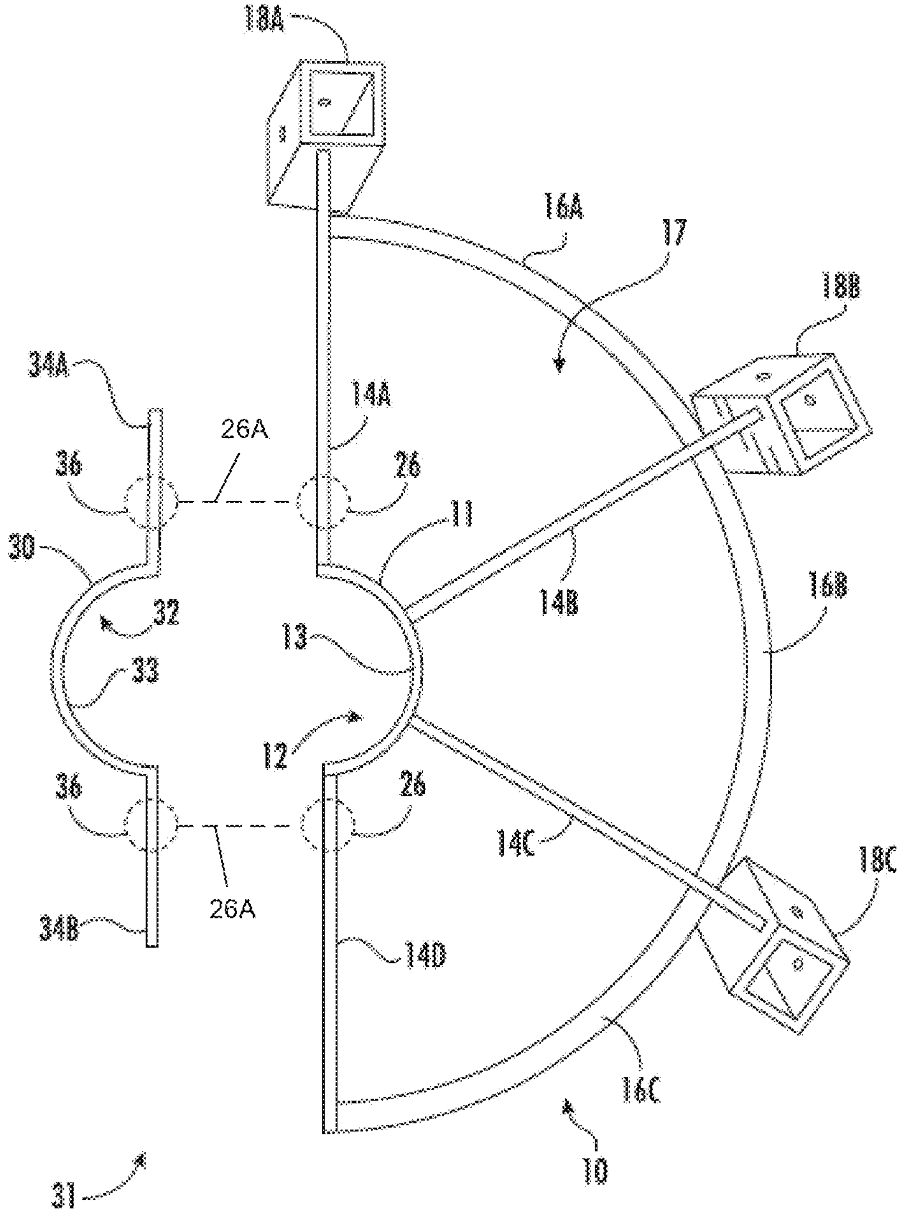

21
9
11
18B
14D
25
11
25
18A
14C
14D
18C
16C
18C
16C
14C
17
16B
18B
14B
14A
16A
18A
21

FIG. 1E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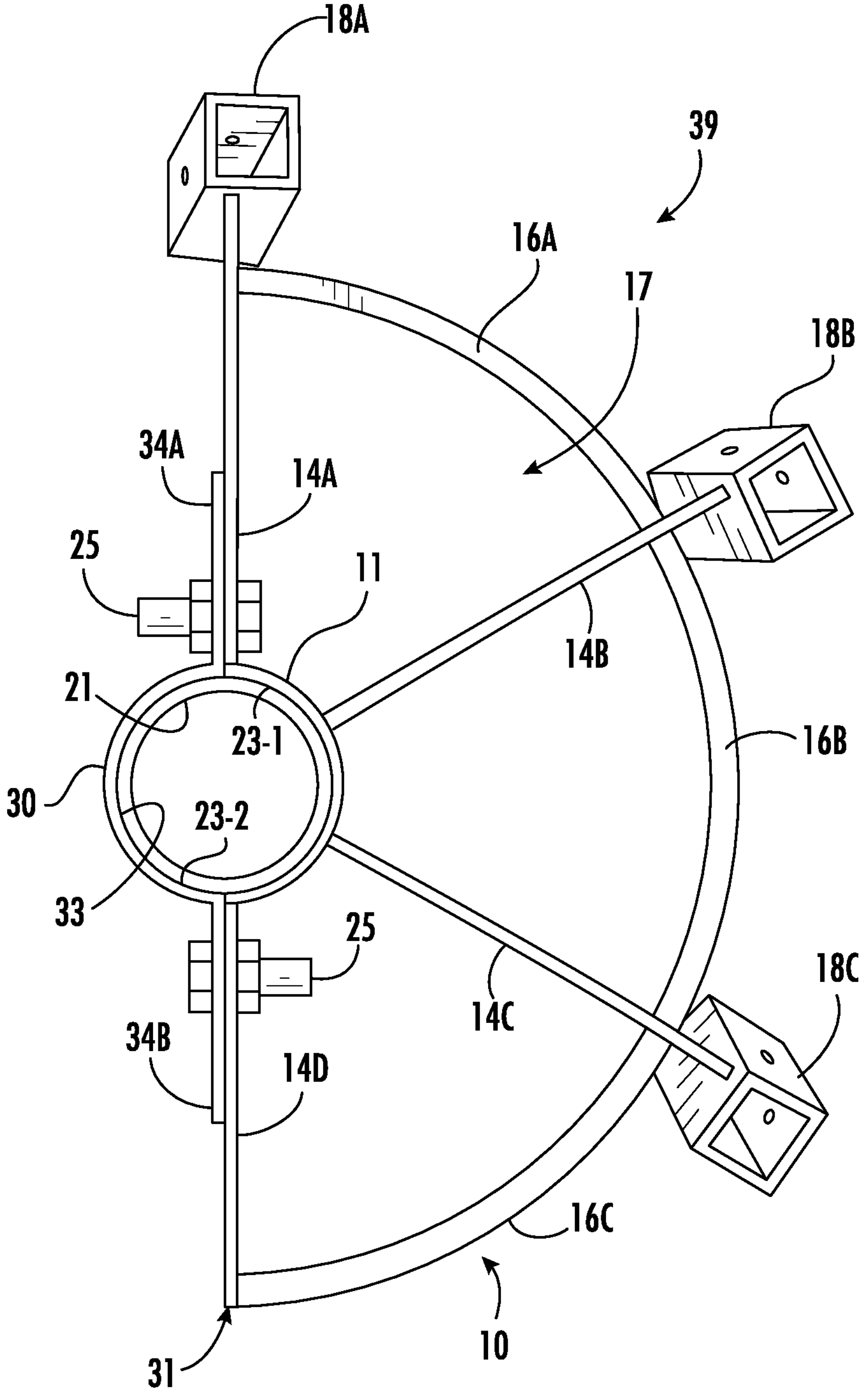
16A
17
18B
34A
26A
14A
36
26
11
30
32
14B
16B
13
33
12
36
26
14C
18C
26A
34B
14D
16C
10
31

FIG. 2A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39
16A
17
18B
34A
14A
25
11
14B
21
23-1
16B
30
23-2
33
25
18C
14C
34B
14D
16C
10
31

FIG. 2B

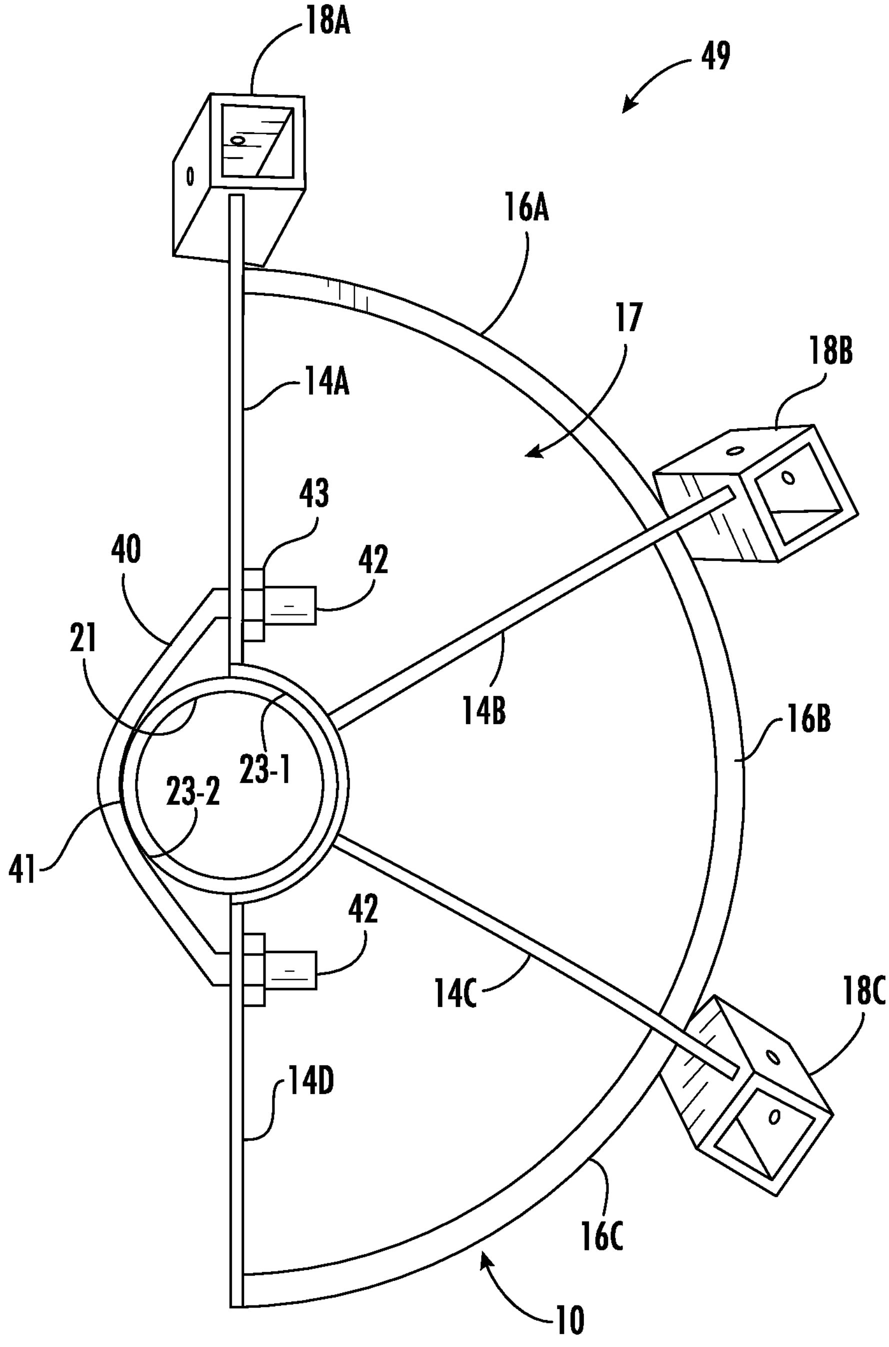
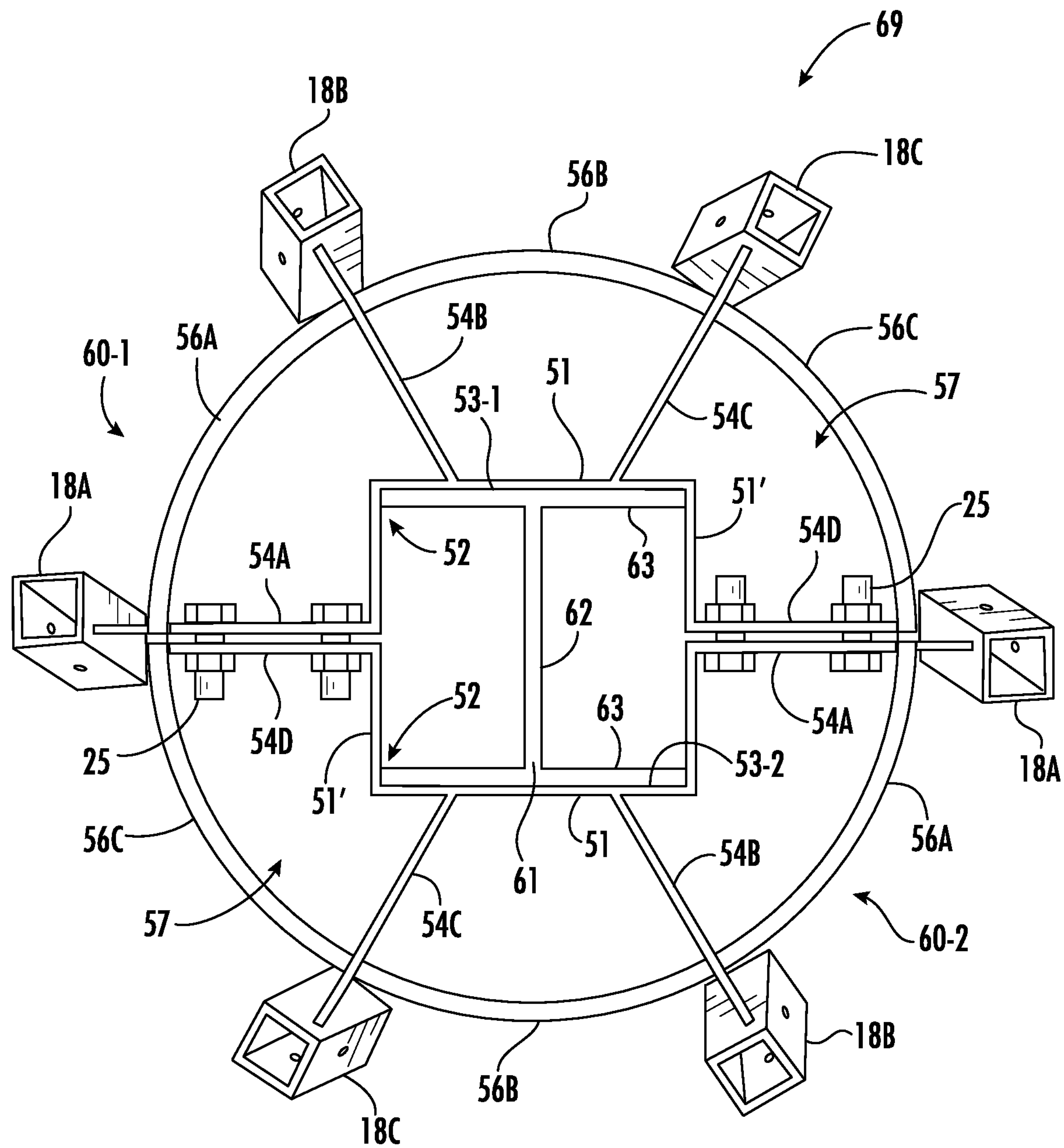

18A
49
16A
17
18B
14A
43
40
42
21
14B
16B
23-1
23-2
41
42
14C
18C
14D
16C
10

FIG. 3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18B
18C
56B
56C
54B
60-1
56A
51
53-1
54C
57
18A
51′
25
54D
54A
52
63
62
25
54D
52
63
54A
18A
51′
53-2
56C
61
51
56A
57
54C
54B
60-2
18B
56B
18C

FIG. 4A

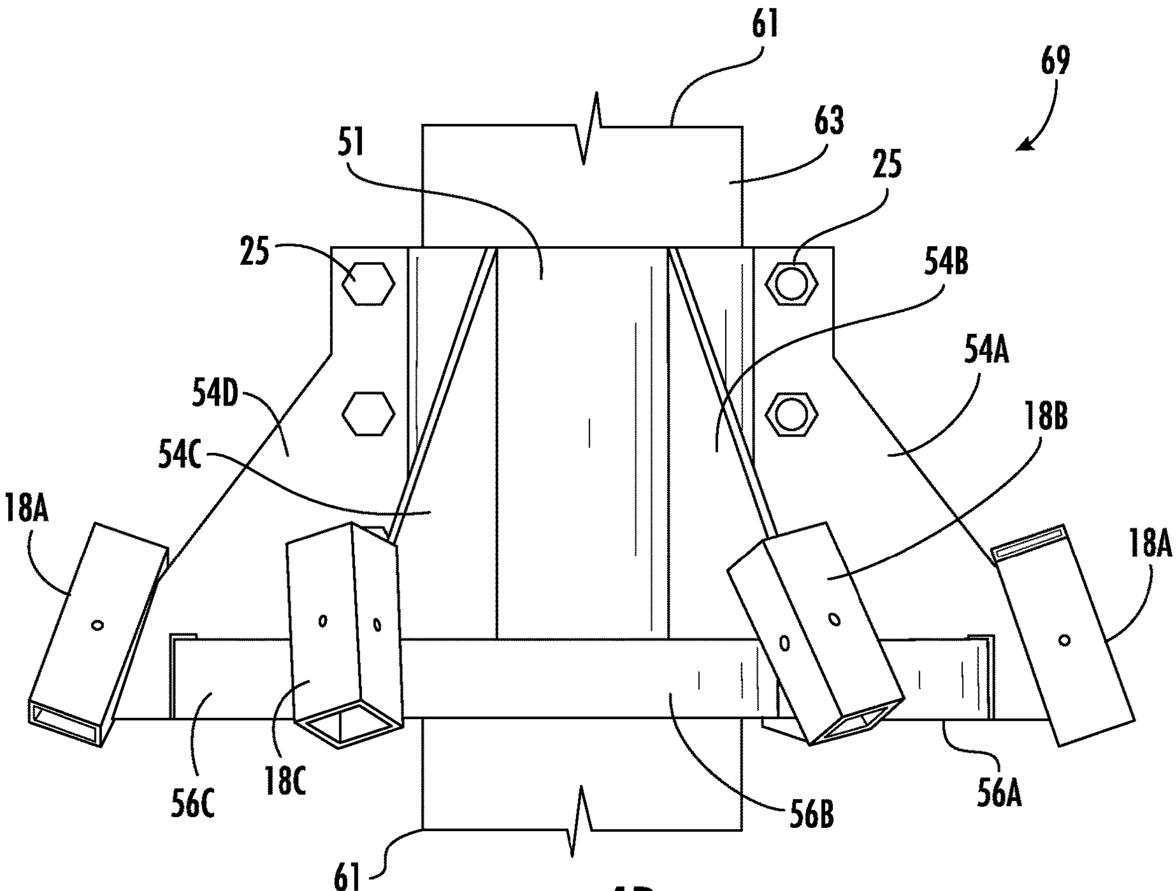

61
69
63
51
25
25
54B
54A
54D
18B
54C
18A
18A
56C
18C
56B
56A
61

FIG. 4B

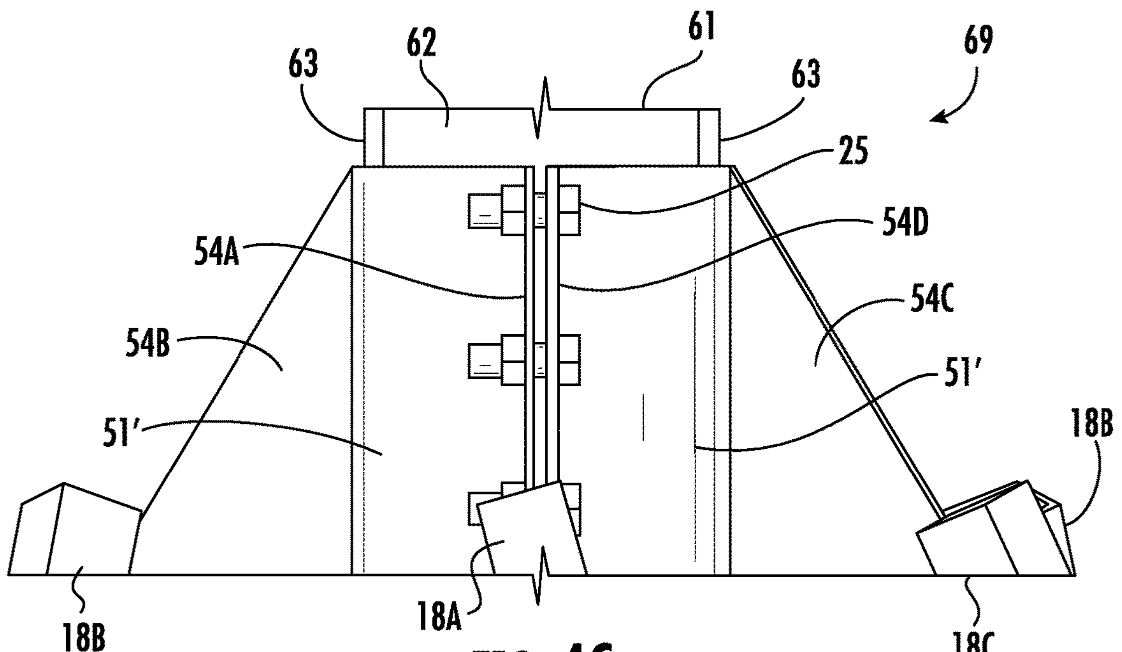

61
63
62
63
69
25
54D
54A
54C
54B
51′
51′
18B
18B
18A
18C

FIG. 4C

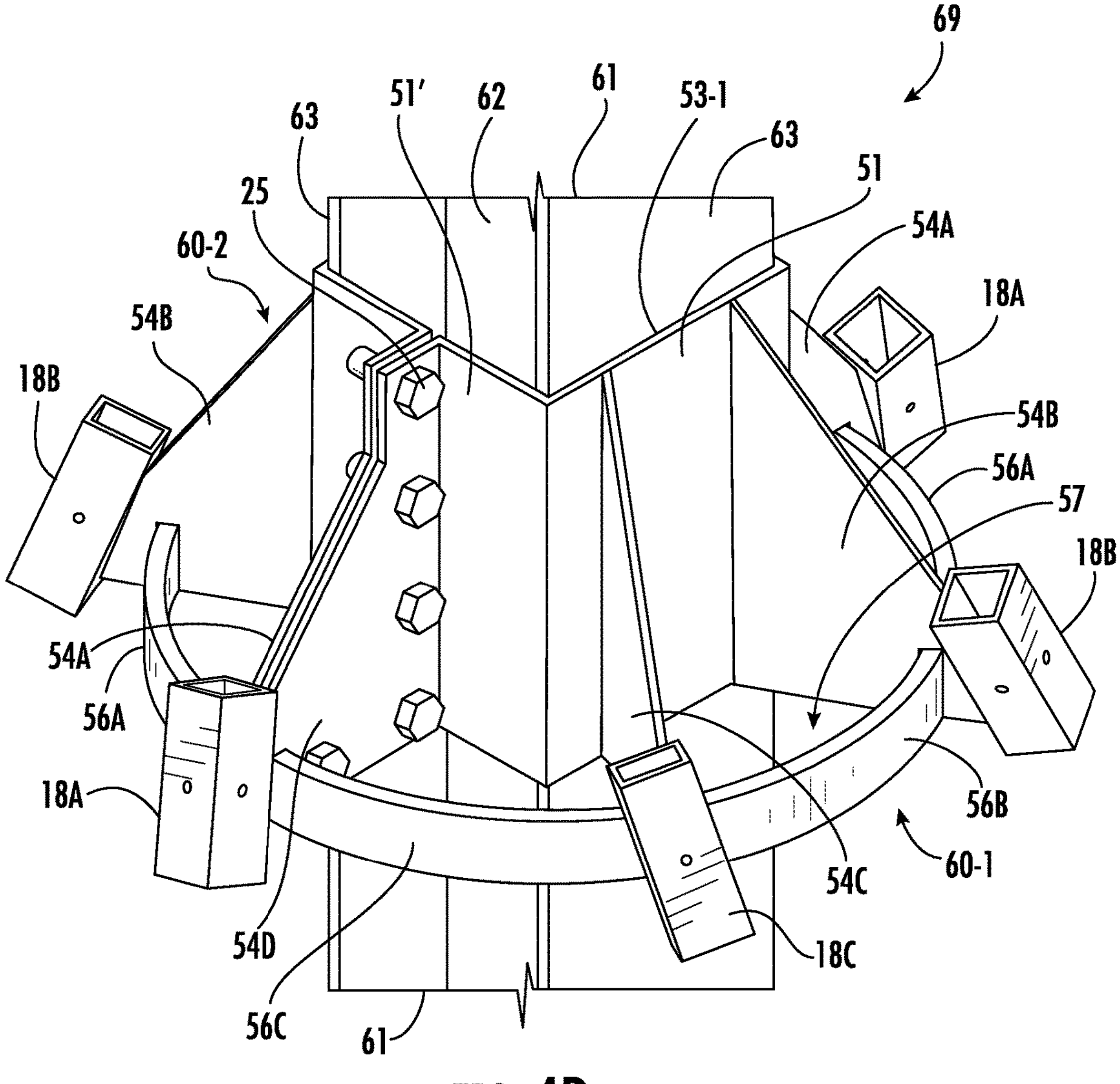

69
63
51′
62
61
53-1
63
51
25
60-2
54A
18A
54B
18B
54B
56A
57
18B
54A
56A
18A
54D
56C
61
18C
54C
60-1
56B

FIG. 4D

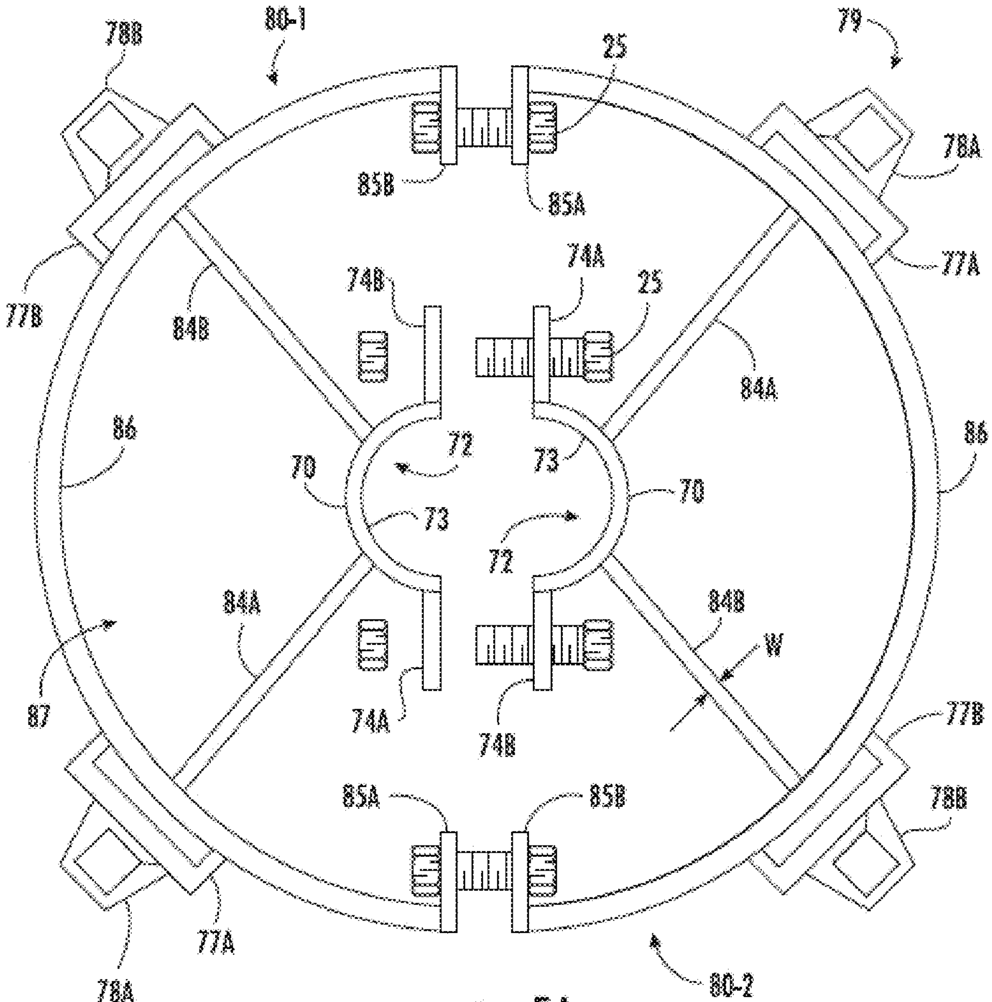

78B
80-1
25
79
78A
85B
85A
77A
74A
74B
25
77B
84B
84A
86
72
73
86
70
70
73
72
84A
84B
W
87
74A
74B
77B
85A
85B
78B
77A
78A
80-2

FIG. 5A

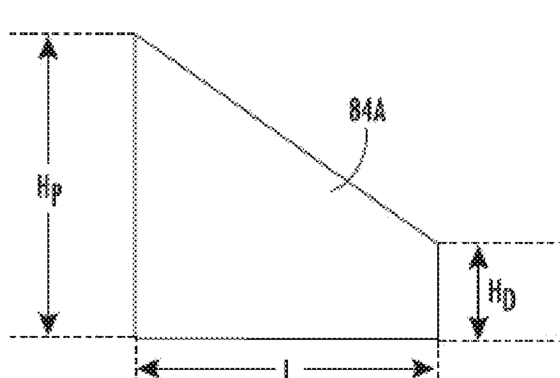

84A
H_P
H_D
L

FIG. 5B

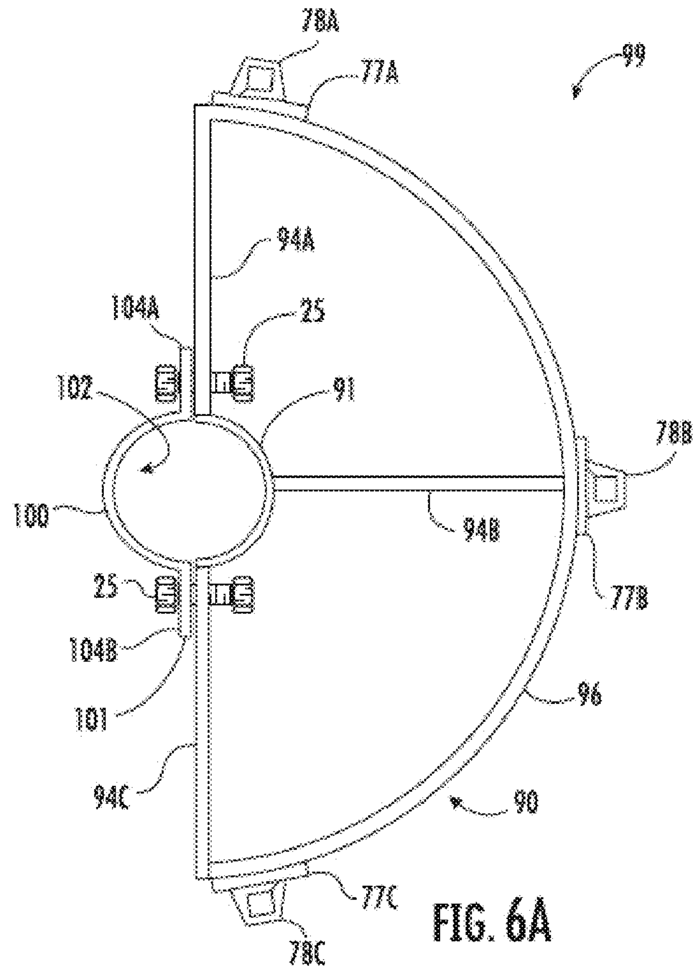


FIG. 6A

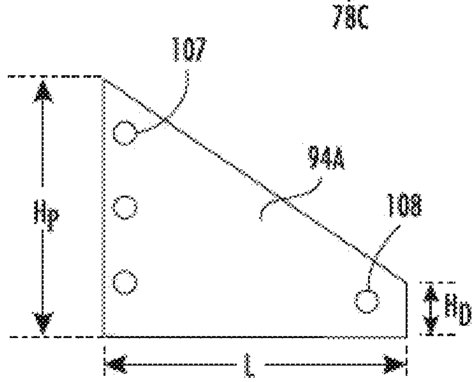


FIG. 6B

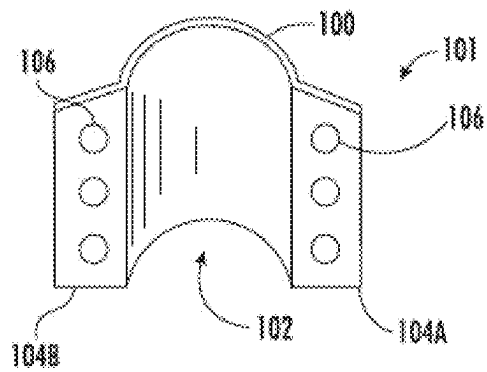
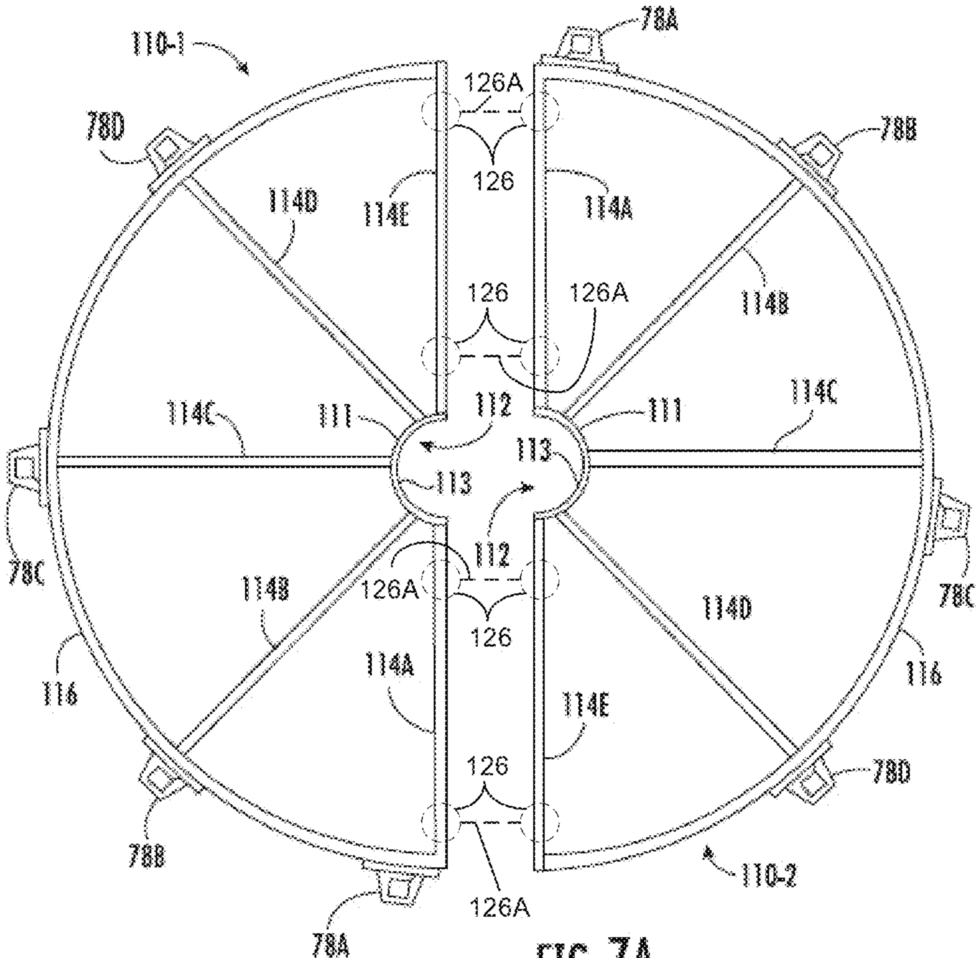


FIG. 6C


110-1
78A
78D
126A
78B
126
114D
114E
114A
126
126A
114B
114C
111
112
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114C
113
113
78C
112
114B
126A
126
114D
78C
114A
116
116
114E
126
78D
78B
110-2
126A
78A

FIG. 7A

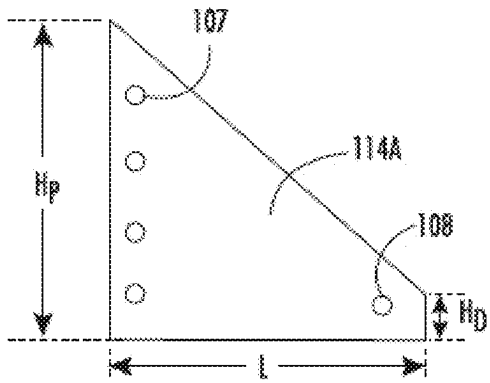

107
114A
Hp
108
H_D
L

FIG. 7B

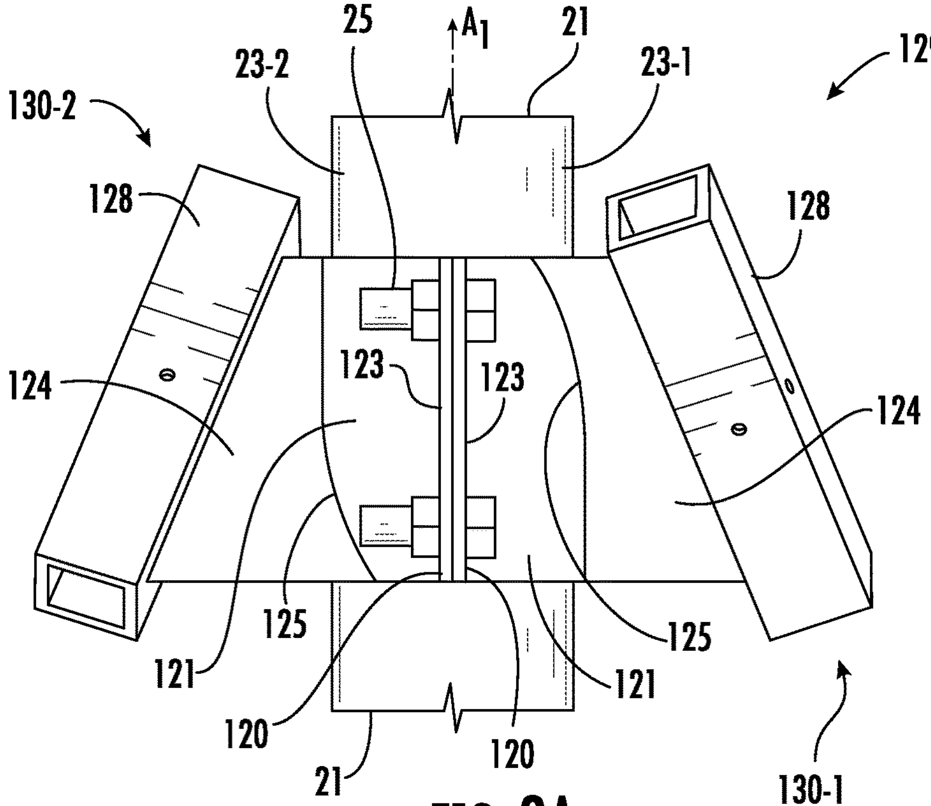

25
A1
21
23-2
23-1
129
130-2
128
128
124
123
123
124
125
125
121
121
120
120
21
130-1

FIG. 8A

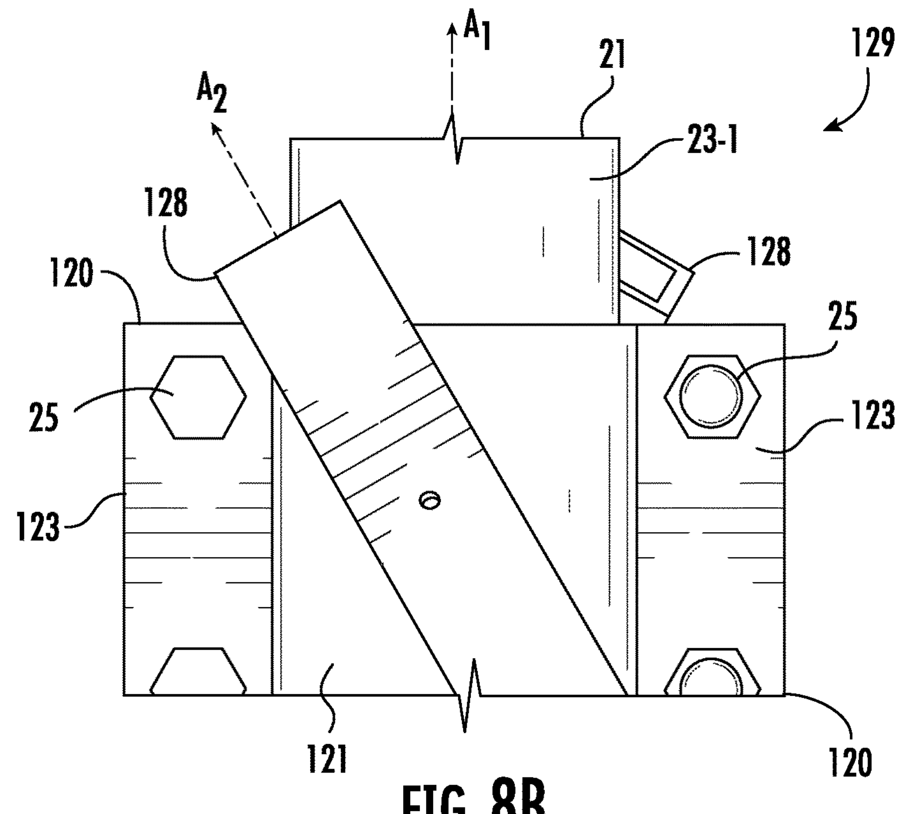
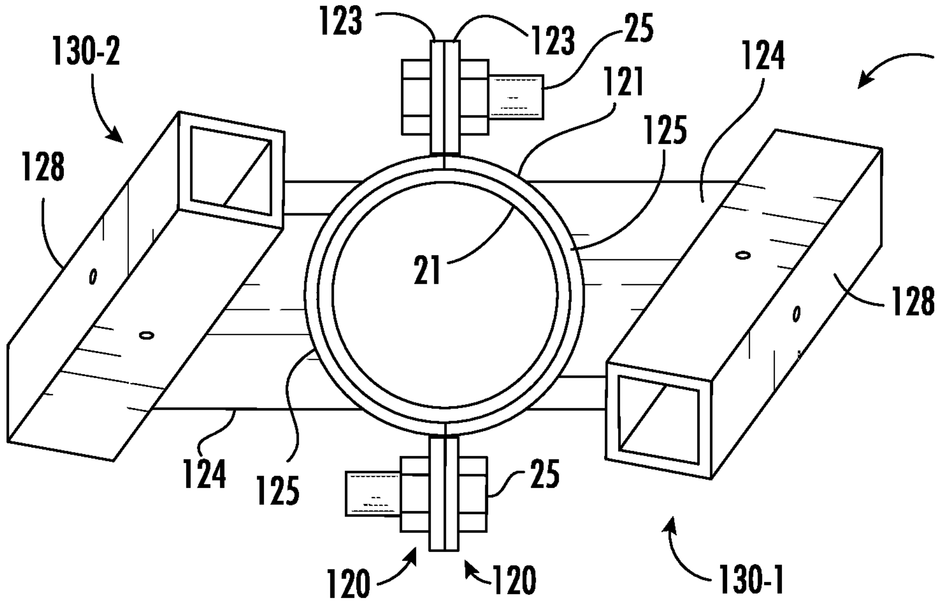

A1
21
129
A2
23-1
128
128
120
25
25
123
123
121
120

FIG. 8B


123
123
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130-2
121
124
129
125
128
21
128
124
125
25
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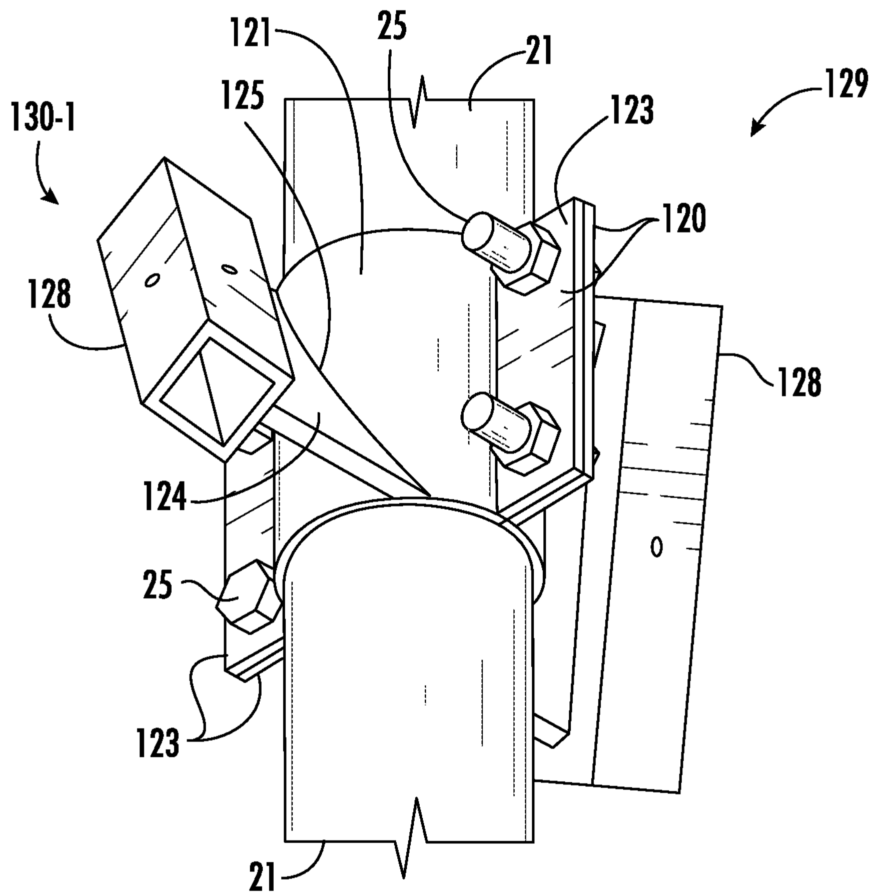
120
130-1

FIG. 8C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130-1
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128
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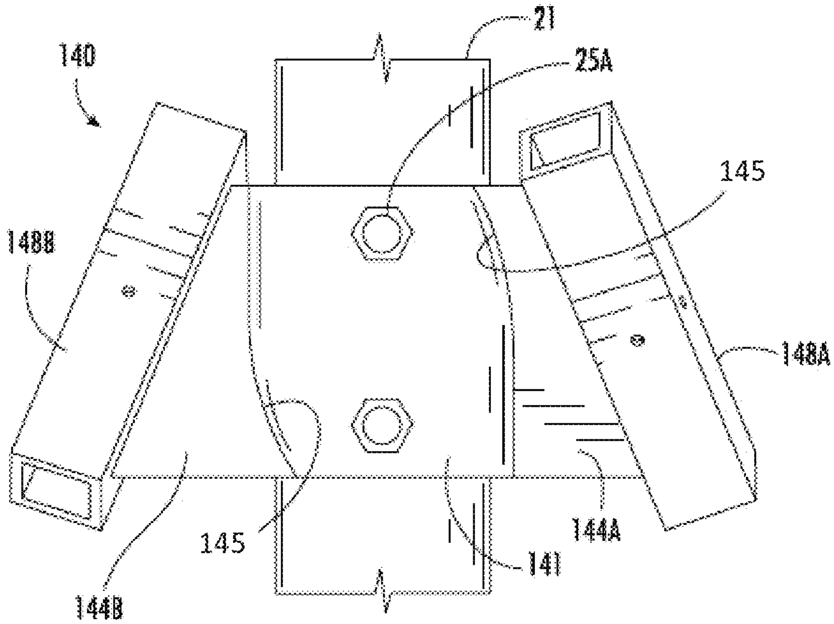
25
123
21

FIG. 8D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21
25A
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148B
148A
145
144A
141
144B

FIG. 9A

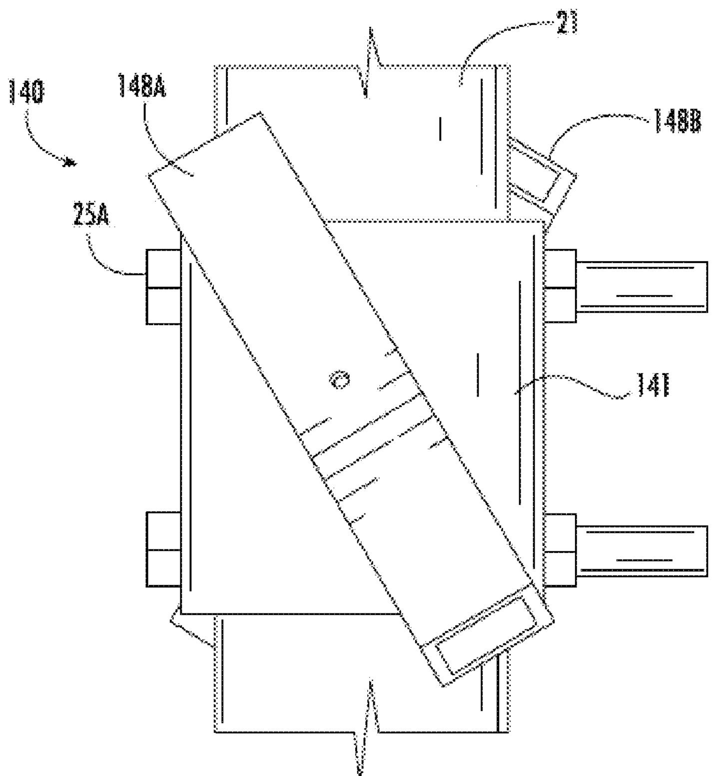

21
140
148A
148B
25A
141

FIG. 9B

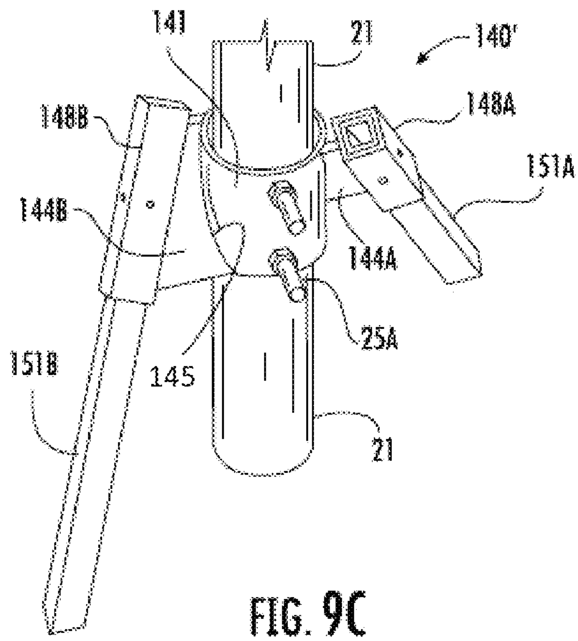
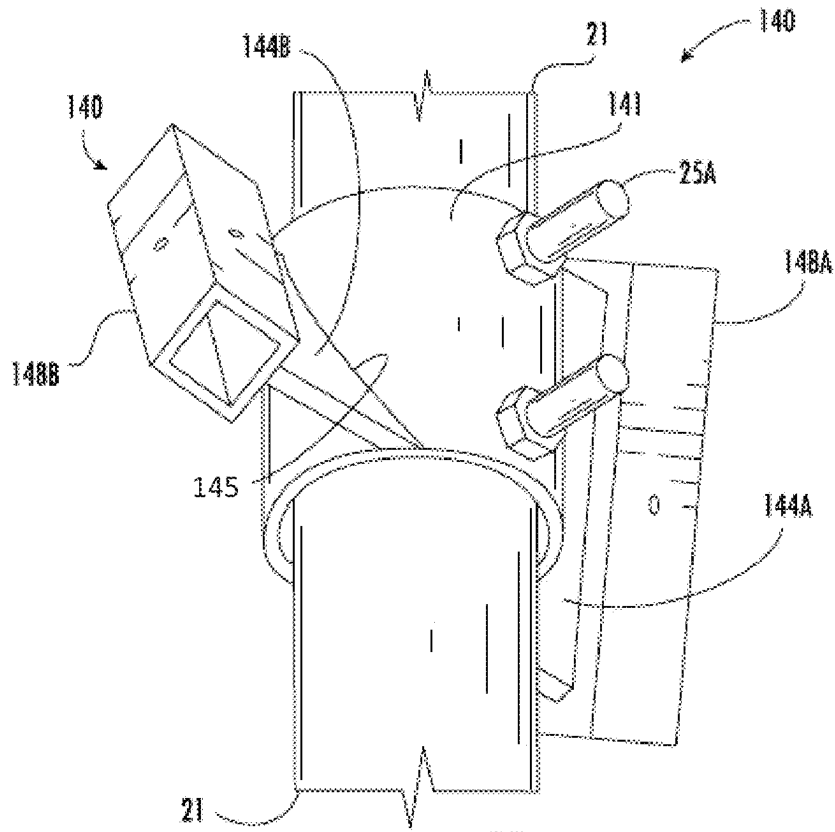

141
21
140'
148B
148A
151A
144B
144A
25A
145
151B
21

FIG. 9C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148B
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144A
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FIG. 9D

PILE CONNECTION APPARATUS AND METHOD FOR SUPPORTING A VERTICAL PILE

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims priority to U.S. Provisional Patent Application No. 63/312,223 filed on Feb. 21, 2022, wherein the entire contents of the foregoing application are hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

The disclosure relates to a pile connection apparatus and method for supporting a vertical pile, such as may be useful to avoid the need for forming concrete foundations or other foundation types, for the same purpose.

BACKGROUND

Installations of many types (e.g., for streetlights, parking lot lights, fences, display signs, solar panels, and the like) may require a ground-contacting support structure. Such installations may require vertically driven piles that are driven into the ground to a specified depth to provide sufficient support. In many installations, piles are embedded in or attached to concrete foundations, either precast or poured on-site, for additional support. While concrete foundations utilize inexpensive materials, their installation may require significant equipment, labor, and time, thereby increasing their overall expense. In climates that are subject to freezing precipitation, exposed concrete foundations may also be vulnerable to ingress of liquid around surfaces thereof followed by freezing, and repeated freeze-thaw cycles with concomitant ice expansion may cause concrete foundations to rise relative to a ground level, thereby reducing their ability to support piles projecting upward therefrom.

Sometimes, a pile is driven to only a portion of a desired depth but then hits an obstruction, resulting in a refusal or rejection. A refusal refers to an inability for a pile (which may also be referred to as a beam) to reach a desired depth to maintain stability of a support structure. The obstruction or impediment (e.g., compacted substrates, rocks, foreign objects, etc.) impede driving the pile to the desired depth. In such circumstances, a pile usually cannot be moved to a different location when large integrated equipment is being installed. Accordingly, some installers remove a pile, drill through the obstruction, replace the pile, and pour concrete around the installed pile. Such a process is slow and expensive. Other times, a pre-existing pile may need further structural support, such as due to environmental changes. Processes for addressing such issues are typically slow, labor-intensive, and expensive.

Need therefore exists for improved pile connection apparatuses for supporting new or pre-existing vertical piles, to address limitations associated with piles supported by conventional foundations or otherwise driven into ground.

SUMMARY

Aspects disclosed herein include a pile connection apparatus that comprises at least one fixture having a collar portion, a plurality of micropile sleeves, a plurality of gusset members extending between the collar portion and the micropile sleeves, and at least one connecting member coupling adjacent gusset members and being spaced apart from the collar portion. A recess (which in certain embodiments may surround approximately one-half of a perimeter or projected top area of a vertical pile) as defined by the collar portion is bounded by a wall of the collar portion configured to contact a portion of a vertical pile to add support thereto. One or more fasteners may be used to press the wall of the collar portion against the vertical pile, with certain embodiments employing two complementary fixtures that may be coupled to one another (e.g., using the fasteners) along opposing sides of the vertical pile. Each micropile sleeve is configured to receive a micropile that may be driven through the sleeve into the ground at a position radially or laterally spaced apart from the vertical pile.

In one aspect, the disclosure relates to a pile connection apparatus for supporting a vertical pile that comprises a first fixture comprising a first collar portion, a plurality of first micropile sleeves, a plurality of first gusset members coupling the plurality of first micropile sleeves to the first collar portion, and at least one first connecting member coupling adjacent first gusset members of the plurality of first gusset members. The first collar portion comprises a first recess configured to receive a first portion of the vertical pile, with the first recess being bounded by a first wall configured to contact a first pile surface of the first portion of the vertical pile. The plurality of first gusset members project laterally outward from the first collar portion, and the at least one first connecting member is spaced apart from the first collar portion. One or more of (i) the first collar portion and (ii) at least one first gusset member, either comprises, or is configured to receive, at least one fastener configured to cooperate with the one or more of (i) the first collar portion and (ii) at least one first gusset member, and cause the first wall to engage the first pile surface.

In certain embodiments, the plurality of first gusset members are intermediately coupled to the plurality of first micropile sleeves, with the at least one first connecting member arranged between (a) the plurality of first gusset members and (b) the plurality of first sleeve micropile sleeves.

In certain embodiments, each first gusset member of the plurality of first gusset members comprises: (i) a length extending in a direction from the first collar portion to a corresponding first micropile sleeve of the plurality of first micropile sleeves, (ii) a width orthogonal to the length, (iii) a proximal height at a point of contact with the first collar portion, and (iv) a distal height at a point of contact with the at least one first connecting member; wherein, for each first gusset member, the proximal height is greater than the distal height, and the distal height is greater than the width.

In certain embodiments, each first micropile sleeve of the plurality of first micropile sleeves has a first sleeve inlet, a first sleeve outlet, and a first sleeve axis extending through the first sleeve inlet and the first sleeve outlet; and for each first micropile sleeve, the first sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus.

In certain embodiments, each first gusset member of the plurality of first gusset members comprises: a first upper surface, a first lower surface, a first proximal surface proximal to the first collar portion, and a first distal surface distal from the first collar portion. For each first gusset member, a first imaginary plane is definable through the first upper surface, the first lower surface, the first proximal surface, and the first distal surface; and for each first gusset member, the first imaginary plane is non-vertical, and is substantially parallel to the first sleeve axis of the first micropile sleeve coupled to the first gusset member.

In certain embodiments, the first wall is curved in shape, with the pile connection apparatus being configured to support a vertical pile having a cylindrical shape; and the first proximal surface of each first gusset member abuts the first wall, with the first proximal surface being curved in shape to substantially conform to a curvature of the first wall.

In certain embodiments, the pile connection apparatus further comprises a second fixture comprising a second collar portion, a plurality of second micropile sleeves, a plurality of second gusset members coupling the plurality of second micropile sleeves to the second collar portion, and at least one second connecting member coupling adjacent second gusset members of the plurality of second gusset members. The second collar portion comprises a second recess configured to receive a second portion of the vertical pile, the second recess being bounded by a second wall configured to contact a second pile surface of the second portion of the vertical pile, wherein the second portion of the vertical pile opposes the first portion of the vertical pile. The plurality of second gusset members project laterally outward from the second collar portion, and the at least one second connecting member is spaced apart from the second collar portion. One or more of (iii) the second collar portion and (iv) the at least one second gusset member, comprises, or is configured to receive, the at least one fastener, and the at least one fastener is configured to cooperate with the one or more of (iii) the second collar portion and (iv) the at least one second gusset member, and cause the second wall to engage the second pile surface, with the second recess of the second fixture opposing the first recess of the first fixture.

In certain embodiments, each second micropile sleeve of the plurality of second micropile sleeves has a second sleeve inlet, a second sleeve outlet, and a second sleeve axis extending through the second sleeve inlet and the second sleeve outlet; and for each second micropile sleeve, the second sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus.

In certain embodiments, the first recess is configured to receive the first portion of the vertical pile extending around a first perimetral portion spanning a range of 40 to 50 percent of a perimeter or projected top area of the vertical pile, and the second recess is configured to receive the second perimetral portion of the vertical pile extending around a second perimetral portion spanning a range of 40 to 50 percent of the perimeter or a projected top area of the vertical pile.

In certain embodiments, a plurality of first fastener openings are defined in one or more of (i) the first collar portion and (ii) the at least one first gusset member, a plurality of second fastener openings is defined in one or more of (iii) the second collar portion and (iv) the at least one second gusset member, and wherein the first fastener openings are configured to be aligned with the second fastener openings to receive fasteners extending through the aligned first and second fastener openings.

In certain embodiments, one or more fasteners of the at least one fastener is affixed to or integrated with one or more of (i) the first collar portion, (ii) at least one first gusset member, (iii) the second collar portion, and (iv) the at least one second gusset member.

In certain embodiments, the plurality of first gusset members comprises at least three first gusset members.

In certain embodiments, one or more of (i) the first collar portion and (ii) the at least one first gusset member, comprises fastener openings configured to receive at least one U-bolt that is configured to extend around a portion of a perimeter of the vertical pile.

In another aspect, the disclosure relates to a pile connection apparatus for supporting a vertical pile, the apparatus comprising a first fixture, a second fixture, and a plurality of fasteners. The first fixture comprises a first collar portion, a plurality of first micropile sleeves, a plurality of first gusset members coupling the plurality of first micropile sleeves to the first collar portion, and at least one first connecting member coupling adjacent first gusset members of the plurality of first gusset members. The first collar portion comprises a first recess configured to receive a first portion of the vertical pile, the first recess being bounded by a first wall configured to contact a first pile surface of the first portion of the vertical pile. The plurality of first gusset members project laterally outward from the first collar portion, and the at least one first connecting member is spaced apart from the first collar portion. The second fixture comprises a second collar portion, a plurality of second micropile sleeves, a plurality of second gusset members coupling the plurality of second micropile sleeves to the second collar portion, and at least one second connecting member coupling adjacent second gusset members of the plurality of second gusset members. The second collar portion comprises a second recess configured to receive a second portion of the vertical pile, the second recess being bounded by a second wall configured to contact a second pile surface of the second portion of the vertical pile. The plurality of second gusset members project laterally outward from the second collar portion, and the at least one second connecting member is spaced apart from the second collar portion. The plurality of fasteners is configured to cooperate with the first fixture and the second fixture to cause the first wall to engage the first pile surface, to cause the second wall to engage the second pile surface, and to cause the vertical pile to be frictionally engaged between the first fixture and the second fixture.

In certain embodiments, the plurality of first gusset members are intermediately coupled to the plurality of first micropile sleeves, with the at least one first connecting member arranged between (a) the plurality of first gusset members and (b) the plurality of first sleeve micropile sleeves; and the plurality of second gusset members are intermediately coupled to the plurality of second micropile sleeves, with the at least one second connecting member arranged between (c) the plurality of second gusset members and (d) the plurality of second micropile sleeves.

In certain embodiments, each first gusset member of the plurality of first gusset members comprises: (i) a first length extending in a direction from the first collar portion to a corresponding first micropile sleeve of the plurality of first micropile sleeves, (ii) a first width orthogonal to the length, (iii) a first proximal height at a point of contact with the first collar portion, and (iv) a first distal height at a point of contact with the at least one first connecting member, wherein for each first gusset member, the first proximal height is greater than the first distal height, and the first distal height is greater than the first width. Each second gusset member of the plurality of second gusset members comprises: (i) a second length extending in a direction from the second collar portion to a corresponding second micropile sleeve of the plurality of second micropile sleeves, (ii) a second width orthogonal to the length, (iii) a second proximal height at a point of contact with the second collar portion, and (iv) a second distal height at a point of contact with the at least one second connecting member; wherein for each second gusset member, the second proximal height is greater than the second distal height, and the second distal height is greater than the second width.

In certain embodiments, each first micropile sleeve of the plurality of first micropile sleeves has a first sleeve inlet, a first sleeve outlet, and a first sleeve axis extending through the first sleeve inlet and the first sleeve outlet; wherein, for each first micropile sleeve, the first sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus. Each second micropile sleeve of the plurality of second micropile sleeves has a second sleeve inlet, a second sleeve outlet, and a second sleeve axis extending through the second sleeve inlet and the second sleeve outlet; wherein, for each second micropile sleeve, the second sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus.

In certain embodiments, each first gusset member of the plurality of first gusset members comprises: a first upper surface, a first lower surface, a first proximal surface proximal to the first collar portion, and a first distal surface distal from the first collar portion; and each second gusset member of the plurality of second gusset members comprises: a second upper surface, a second lower surface, a second proximal surface proximal to the second collar portion, and a second distal surface distal from the second collar portion. For each first gusset member, a first imaginary plane is definable through the first upper surface, the first lower surface, the first proximal surface, and the first distal surface; and for each first gusset member, the first imaginary plane is non-vertical, and is substantially parallel to the first sleeve axis of the first micropile sleeve coupled to the first gusset member. For each second gusset member, a second imaginary plane is definable through the second upper surface, the second lower surface, the second proximal surface, and the second distal surface; and for each second gusset member, the second imaginary plane is non-vertical, and is substantially parallel to the second sleeve axis of the second micropile sleeve coupled to the second gusset member.

In certain embodiments, the first wall is curved in shape, with the pile connection apparatus being configured to support a vertical pile having a cylindrical shape; the first proximal surface of each first gusset member abuts the first wall, with the first proximal surface being curved in shape to substantially conform to a curvature of the first wall; the second wall is curved in shape, with the pile connection apparatus being configured to support the vertical pile having a cylindrical shape; and the second proximal surface of each second gusset member abuts the second wall, with the second proximal surface being curved in shape to substantially conform to a curvature of the second wall.

In certain embodiments, the vertical pile configured to be supported by the pile connection apparatus comprises a pile cross-sectional width or diameter; the first recess comprises a maximum horizontal depth with a dimension in a range of 40 to 50 percent of the pile width or diameter; and the second recess comprises a maximum horizontal depth with a dimension in a range of 40 to 50 percent of the pile width or diameter.

In certain embodiments, a plurality of first fastener openings are defined in one or more of (i) the first collar portion and (ii) the at least one first gusset member, a plurality of second fastener openings is defined in one or more of (iii) the second collar portion and (iv) the at least one second gusset member, and wherein the first fastener openings are configured to be aligned with the second fastener openings to receive fasteners extending through the aligned first and second fastener openings.

In certain embodiments, one or more fasteners of the plurality of fasteners is affixed to or integrated with one or more of (i) the first collar portion, (ii) at least one first gusset member, (iii) the second collar portion, and (iv) the at least one second gusset member.

In another aspect, the disclosure relates to a method for supporting a vertical pile, the method comprising: positioning a pile vertically to a ground to form a vertical pile; coupling a pile connection apparatus as disclosed herein to the vertical pile, with the first collar portion of the first fixture receiving a first portion of the vertical pile, with the second collar portion of the second fixture receiving a second portion of the vertical pile, and with the plurality of fasteners cooperating with the first and second collar portions to cause the vertical pile to be frictionally engaged therebetween; inserting a plurality of first micropiles through the plurality of first micropile sleeves and into the ground; and inserting a plurality of second micropiles through the plurality of second micropile sleeves and into the ground.

In another aspect, any of the foregoing aspects or embodiments disclosed herein may be combined for additional advantage. Moreover, it is specifically contemplated that features of any one or more embodiments disclosed herein may be combined with or substituted for one another, unless indicated to the contrary.

It is to be understood that both the foregoing general description and the following detailed description are merely exemplary and are intended to provide an overview or framework for understanding the nature and character of the claims. The accompanying drawings are included to provide a further understanding and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiment(s) and, together with the description, serve to explain principles and operation of the various embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is a top plan view of complementary first and second fixtures that may be coupled to one another to form a pile connection apparatus according to one embodiment, with each fixture including a recess-defining collar portion, three micropile sleeves, multiple gusset members projecting laterally outward (i.e., toward the micropile sleeves), and connecting members coupling adjacent gusset members, with the connecting members being laterally spaced apart from the collar portion to provide gaps therebetween.

FIG. **1**B is a top plan view of the first and second fixtures of FIG. **1**A being joined together with fasteners to form a pile connection apparatus according to one embodiment in which a generally cylindrical vertical pile is contacted (and supported) by walls bounding the recesses of the collar portions.

FIG. **1**C is a side elevational view of the pile connection apparatus of FIG. **1**B.

FIG. **1**D is a front elevational view of the pile connection apparatus of FIG. **1**B.

FIG. **1**E is a perspective view of the pile connection apparatus of FIG. **1**B.

FIG. **2**A is a top plan view of a fixture according to FIG. **1**A positioned adjacent to a coupling member that defines a collar portion, with the fixture and coupling member being configured to be coupled together with fasteners to form a pile connection apparatus.

FIG. 2B is a top plan view of the fixture and coupling member of FIG. 2A being coupled together with fasteners to form a pile connection apparatus in which a generally cylindrical vertical pile is contacted (and supported) by walls bounding the recesses of the collar portions.

FIG. 3 is a top plan view of a fixture according to FIG. 1A cooperating with a U-bolt to form a pile connection apparatus in which a generally cylindrical vertical pile is contacted (and supported) between the U-bolt and a wall bounding a recesses of the collar portion of the fixture.

FIG. 4A is a top plan view of first and second fixtures (each similar to the figures shown in FIG. 1A but defining rectangular recesses in collar portions thereof) being joined together with fasteners to form a pile connection apparatus according to one embodiment in which a vertical pile having a rectangular cross-sectional shape is contacted (and supported) by walls bounding the recesses of the collar portions.

FIG. 4B is a front elevational view of the pile connection apparatus of FIG. 4A.

FIG. 4C is a front elevational view of the pile connection apparatus of FIG. 4A.

FIG. 4D is a perspective view of the pile connection apparatus of FIG. 4A.

FIG. 5A is a top plan view of first and second fixtures configured to be coupled together with fasteners to form a pile connection apparatus according to one embodiment, with each fixture including two gusset members, a single connecting member connecting the gusset members, a collar portion, two micropile sleeves indirectly coupled to the gusset members through the connecting member, and fastener receiving holes defined in outwardly projecting tab extensions of a collar portion and defined in inwardly projecting tab extensions of the connecting member.

FIG. 5B is a side elevational view of a gusset member of a fixture of FIG. 5A.

FIG. 6A is a top plan view of a fixture joined by fasteners to a coupling member to form a pile connection apparatus according to one embodiment, the fixture including three gusset members, a single connecting member connecting the gusset members, three micropile sleeves indirectly coupled to the gusset members through the connecting member, and a collar portion, with fastener receiving holes defined in two gusset members and in projecting tab extensions of the coupling member.

FIG. 6B is a side elevational view of a gusset member of the fixture of FIG. 6A.

FIG. 6C is a perspective view of the coupling member of FIG. 6A.

FIG. 7A is a top plan view of first and second fixtures configured to be coupled together with fasteners to form a pile connection apparatus according to one embodiment, with each fixture including five gusset members, a single connecting member connecting the gusset members, a collar portion, four micropile sleeves indirectly coupled to the gusset members through the connecting member, and fastener receiving holes defined in selected gusset members.

FIG. 7B is a side elevational view of a gusset member of a fixture of FIG. 7A.

FIG. 8A is side elevational view of a pile connection apparatus affixed to a cylindrical vertical pile, the pile connection apparatus including two complementary fixtures joined with fasteners, with each fixture having a collar portion and one gusset member extending from the collar portion to a micropile sleeve without any connecting members spanning between micropile sleeves, with each gusset member being inclined away from vertical and having a curved proximal surface to conform to a curved surface of a wall of the collar portion.

FIG. 8B is a front elevational view of a portion of the pile connection apparatus and vertical pile of FIG. 8A.

FIG. 8C is a top plan view of the pile connection apparatus and vertical pile of FIG. 8A.

FIG. 8D is a lower perspective view of the pile connection apparatus and vertical pile of FIG. 8A.

FIG. 9A is a front elevational view of a pile connection apparatus including a single fixture with a collar fitting around an entire perimeter of a vertical pile and fasteners extending through the collar and the vertical pile, with two opposing gusset members extending from the collar to two micropile sleeves, and with each gusset member being inclined away from vertical and having a curved proximal surface to conform to a curved surface of a wall of the collar portion.

FIG. 9B is a side elevational view of the pile connection apparatus and vertical pile of FIG. 9A.

FIG. 9C is an upper perspective view of the pile connection apparatus and vertical pile of FIG. 9A, with micropiles received by the micropile sleeves.

FIG. 9D is a lower perspective view of the pile connection apparatus and vertical pile of FIG. 9A.

DETAILED DESCRIPTION

Reference will now be made in detail to the presently preferred embodiments, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts.

Terms such as "left," "right," "top," "bottom," "front," "back," "horizontal," "parallel," "perpendicular," "vertical," "lateral," "coplanar," and similar terms are used for convenience of describing the attached figures and are not intended to limit this description. For example, terms such as "left side" and "right side" are used with specific reference to the drawings as illustrated, and the embodiments may be in other orientations in use. Further, as used herein, terms such as "horizontal," "parallel," "perpendicular," "vertical," "lateral," etc., include slight variations that may be present in working examples.

It will be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element without departing from the scope of the present disclosure. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

It will be understood that when an element is referred to as being "connected" or "coupled" to another element, it can be directly connected or coupled to the other element, or intervening elements may be present. In contrast, when an element is referred to as being "directly connected" or "directly coupled" to another element, there are no intervening elements present.

The term "vertical" as used herein, may be generally vertical, such as +/−30 degrees, +/−15 degrees, +/−10 degrees, +/−5 degrees, or +/−3 degrees in certain embodiments.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises," "comprising," "includes," and/or "including" when used herein specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Disclosed is a pile connection apparatus intended to add support to a vertical pile. A pile connection apparatus may utilize one fixture or complementary matching fixtures configured to be coupled with a vertical pile, wherein each fixture includes multiple micropile sleeves each configured to receive a micropile at a location laterally spaced apart from the vertical pile. When one fixture is used, a coupling member and fasteners, or alternatively only fasteners such as U-bolts, may be used to cause a fixture to be coupled with a vertical pile. When multiple fixtures (e.g., two complementary fixtures) are used, such fixtures may be coupled to one another using fasteners (e.g., threaded bolts and nuts, rivets, or other conventional fasteners) around a vertical pile to be supported. Each fixture includes a collar portion that defines a recess configured to receive a portion of a vertical pile. For example, first and second collar portions of complementary fixtures may include walls defining recesses that are semicircular when viewed from above, and when the collar portions are mated together, the recess-defining walls of the respective collar portions may surround substantially an entire perimeter of a portion of a cylindrical pile to be supported. Alternatively, first and second collar portions of complementary fixtures may include walls defining recesses that are rectangular in shape when viewed from above, and when the collar portions are mated together, the recess-defining walls of the respective collar portions may surround substantially an entire perimeter or projected top boundary of a portion of a pile having a rectangular cross-sectional shape. If a single fixture is used in connection with a coupling member, the coupling member may define a recess that may at least partially conform to one half of a perimeter or a projected top boundary of a portion of a vertical pile to be supported, with the fixture defining a complementary recess at least partially conforming to another half of a perimeter of a projected top boundary of the vertical pile.

In certain embodiments, a pile useable with a pile connection apparatus as disclosed herein includes structural metal, such as structural steel, structural aluminum, or other materials. In certain embodiments, the structural metal includes circular tubing, rectangular tubing, square tubing, and flanged bearing piles of various cross-sectional shapes (e.g., I-shaped piles, H-shaped piles, W-shaped piles, C-shaped piles, S-shaped piles, etc., which may also be referred to as beams). In certain embodiments, a pile includes timber (e.g., rectangular cross-section, square cross-section, round cross-section, etc.).

In one embodiment, a pile connection apparatus includes two matching fixtures configured to support (e.g., by compressive engagement) an elongated vertical pile extending along a pile axis. The pile connection apparatus includes a first fixture having a first collar portion defining a recess configured to receive (and preferably match) one side of an elongated pile extending along a pile axis. The first fixture is configured to partially surround the elongated pile. The first fixture includes multiple micropile sleeves coupled with the first collar portion by radially extending first gusset members. Each (first) micropile sleeve comprises a first inlet, a first outlet, and a bore extending therebetween, and is configured to receive a corresponding micropile. A micropile may embody a stake, shaft, or other elongated implement that is generally smaller in length and/or width (e.g., diameter) than a vertical pile to be supported by the pile connection apparatus. The pile connection apparatus further includes a second fixture having a second collar portion configured to receive (and preferably match) an other side of the elongated pile. In certain embodiments, the second fixture may mirror the first fixture. The second fixture further includes multiple micropile sleeves coupled with the second collar portion by radially extending second gusset members. Each (second) micropile sleeve comprises a first inlet, a first outlet, and a bore extending therebetween, and the is configured to receive a corresponding micropile. The first and second fixtures may define holes (e.g., in gusset members, in collar portions (including extending tab portions thereof), or other locations) configured to be aligned with one another to permit the passage of fasteners (e.g., bolts, rivets, etc.) so that the fixtures may be coupled to one another to compressively engage the vertical pile, thus tightly securing the fixtures to the vertical pile.

Fixtures and coupling members as disclosed herein may desirably be formed of metal such as stainless steel, carbon steel, aluminum, or another suitable metal or metal alloy. Since fixtures and coupling members may be positioned outdoors and in contact with a ground surface, such items may have corrosion-resistant features or coatings (e.g., galvanization, paint, and/or other coatings). Material processing methods that may be used for fabricating fixtures and coupling members (and components thereof) may include one or more of rolling, bending, stamping, casting, cutting, machining, welding, and other metal processing steps known in the art.

In certain embodiments, micropile sleeves may be coupled directly to gusset members. In certain embodiments, micropile sleeves may be indirectly coupled to gusset members, with one or more connecting members coupled to the gusset members, and with the micropile sleeves coupled to the one or more connecting members.

One or more connecting members may be used to enhance torsional and structural rigidity of a fixture, which may permit use of longer gusset members (potentially enhancing stability of a pile supported by the pile connection apparatus) and/or permit use of thinner gusset members or less robust mechanical joining (e.g., smaller weld beads) between gusset members and a collar portion. Spacing the connecting members away from a collar portion (e.g., instead of using a continuous plate connecting the collar portion and the gusset members), and providing a gap therebetween, minimizes the potential for accumulation of precipitation, which could facilitate more rapid corrosion and potential for mechanical failure of a fixture.

In certain embodiments, micropile sleeves may include bores configured to guide micropiles in a direction that is not perpendicular to ground, such as an angle in a range of 5 to

35 degrees, or in a range of 10 to 30 degrees, or 15 to 25 degrees, away from true vertical. In certain embodiments, different micropile sleeves may be arranged at different absolute angles, but the same relative angle in comparison to a gusset member proximate to a corresponding micropile sleeve. In certain embodiments, micropile sleeves may include bores configured to guide micropiles in a vertical direction so that micropiles may be driven straight downward into ground.

Reference will now be made to various figures and embodiments disclosed therein.

FIG. **1**A is a top plan view of complementary first and second fixtures **10-1**, **10-2** that may be coupled to one another to form a pile connection apparatus according to one embodiment. Each fixture **10-1**, **10-2** includes a collar portion **11** having a wall **13** defining a semi-cylindrical recess **12** (appearing semi-circular when viewed from above), three micropile sleeves **18**A-**18**C, and multiple gusset members **14**A-**14**D projecting laterally outward (i.e., toward the micropile sleeves **18**A-**18**C). Connecting members **16**A-**16**C that are curved in shape are coupled between adjacent gusset members **14**A-**14**D, with the connecting members **16**A-**16**C being laterally spaced apart from the collar portion **11** to provide gaps **17** between the collar portion and the connecting members **16**A-**16**C, with such gaps **17** being generally wedge-shaped and also extending between adjacent gusset members **14**A-**14**C. The first and second fixtures **10-1**, **10-2** are configured to be coupled together with fasteners (e.g., **25** as shown in FIGS. **1**B-**1**E) that are configured to extend through holes defined in gussets **14**A, **14**D at regions **26**, with dashed lines **26**A representing paths for insertions of such fasteners. Each micropile sleeve **18**A-**18**C defines a bore **19**A-**19**C for receiving a micropile (not shown). As shown, each micropile sleeve **18**A-**18**C is directly coupled to a corresponding gusset member **14**A-**14**C, and each connecting member **16**A-**16**C is directly coupled to an adjacent pair of gusset members **14**A-**14**C, such that a distal portion **14**A'-**14**C' of each gusset member **14**A-**14**C extends beyond the connecting members **16**A-**16**C to contact a corresponding micropile sleeve **18**A-**18**C.

FIGS. **1**B-**1**E provide various views of the first and second fixtures **10-1**, **10-2** of FIG. **1**A being joined together with fasteners **25** (optionally embodied in threaded bolts and nuts, rivets, or the like) to form a pile connection apparatus **9** according to one embodiment in which a generally cylindrical vertical pile **21** is contacted (and supported) by the walls **13** bounding the recesses **12** (shown in FIG. **1**A) of the collar portions **11**. As shown in FIG. **1**B, the fasteners **25** extend through aligned holes (not shown) defined in the first and fourth gusset members **14**A, **14**D of the first and second fixtures **10-1**, **10-2**. Additionally, the vertical pile **21** has two laterally opposing pile surfaces **23-1**, **23-2** that are contacted by walls **13** of the collar portions **11** of the second and first fixtures **10-2**, **10-1**, respectively. In certain embodiments, each collar portion **11** surrounds 50% of a perimeter or projected top border area of the pile **21**, such that both collar portions **11** in combination surround an entire perimeter or projected top border area of the pile **21**. In certain embodiments, each collar portion surrounds a majority (e.g., in a range of 40 to 50 percent) of a perimeter or projected top border area of the pile **21**.

Referring to FIG. **1**C, the gusset members **14**A-**14**D may have a generally triangular shape, with an increased height proximate to the collar portions **11** and a reduced height proximate to the micropile sleeves **18**A-**18**C. Each micropile sleeve (but illustrated only in connection with **18**A) has a sleeve axis A_2, and has a corresponding inlet **28**A and outlet **28**B communicating with the bore thereof (e.g., bore **19**A shown in FIG. **1**A) to permit passage of a micropile. Although each micropile sleeve **18**A-**18**C may have a sleeve axis A_2 pointing in a different absolute direction, each sleeve axis A_2 is non-parallel to a pile axis A_1 of the pile **21**, and each sleeve axis A_2 may have the same or different orientation relative to a corresponding gusset member **14**A-**14**C with which the sleeve member **18**A-**18**C is coupled. FIG. **1**C also shows a micropile **8** extending through one micropile sleeve **18**A, although it is to be appreciated that in use, each micropile sleeve **18**A-**18**C would have a micropile extending therethrough and into the ground (not shown) below the connecting members **16**A-**16**C. FIG. **1**D shows opposing pile surfaces **23-1**, **23-2** at a level above the collar portions **11** of the respective fixtures **10-1**, **10-2**. FIG. **1**E shows a lower portion of the vertical pile **21** extending below the collar portions **11** as being visible through the gaps **17** between the connecting members **16**A-**16**C and the gusset members **14**A-**14**C, but it is to be appreciated that in use, the connecting members **16**A-**16**C may be arranged at a ground surface, with a lower portion of the pile **21** being driven downward into the ground and therefore obscured from view. The remaining elements of FIGS. **1**B to **1**E are the same as previously described in connection with FIG. **1**A. In use, a pile may be positioned vertically relative to a ground (and optionally, driven partially into the ground) to form a vertical pile **21**; the fixtures **10-1**, **10-2** may be coupled (using fasteners **25**) to the vertical pile; and multiple micropiles may be inserted through the micropile sleeves **18**A-**18**C into the ground. In certain embodiments, the fixtures **10-2**, **10-2** may be coupled to the vertical pile **21** after the vertical pile **21** is driven into the ground. In certain embodiments, the fixtures **10-1**, **10-2** may be coupled to the vertical pile **21** before the vertical pile **21** is driven into the ground.

FIGS. **2**A-**2**B illustrate an embodiment in which a single fixture **10** (identical to the fixtures **10-1**, **10-2** of FIG. **1**A) may be used in conjunction with a coupling member **31** and fasteners (**25** in FIG. **2**B) to support a vertical pile. FIG. **2**A shows the fixture **10** positioned adjacent to the coupling member **31** incorporating a collar portion **30** having a wall **33** that defines a recess **32** and having two outwardly projecting tab extensions **34**A, **34**B. The tab extensions **34**A, **34**B define holes at regions **36**, with such holes being aligned with corresponding holes defined in the first and fourth gusset members **14**A, **14**D at regions **26** of the fixture **10** with dashed lines **26**A representing paths for insertion of fasteners (e.g., **25** as shown in FIG. **2**B). The remaining elements of the fixture **10** are the same as those of fixtures **10-1**, **10-2** described in connection with FIG. **1**A. FIG. **2**B shows the fixture **10** and coupling member **31** of FIG. **2**A being coupled together with fasteners **25** (extending through the above-described holes in the tab extensions **34**A, **34**B and in the first and fourth gusset members **14**A, **14**D) to form a pile connection apparatus **39** in which a generally cylindrical vertical pile **21** is contacted (and supported) by walls **13**, **33** bounding the recesses **32**, **12** of the collar portions **11**. In particular, the vertical pile **21** has one pile surface **23-1** contacting the wall **13** of the collar portion **11** of the fixture **10**, and the vertical pile has another, opposing pile surface **23-1** contacting the wall **33** of the collar portion **30** of the coupling member **31**. The pile connection apparatus **39** may be useful in situations where limited support for a pile **21** is needed and/or where lateral clearance is limited (e.g., if the pile **21** is located adjacent to a pre-existing structure such as a wall that would not accommodate presence of a second fixture coupled to the fixture **10**).

As an alternative to using a coupling member or a second fixture, in certain embodiments a fixture may be used with one or more U-bolts to provide support to a vertical pile.

FIG. **3** is a top plan view of a fixture **10** according to FIG. **1**A cooperating with a U-bolt **40** to form a pile connection apparatus **49** in which a generally cylindrical vertical pile **21** is contacted (and supported) between a medial surface **41** of the U-bolt **40** and a wall **13** bounding a recess of the collar portion **11** of the fixture **10**. The U-bolt has threaded ends **42** that cooperate with nuts **43** and extend through holes defined in the first and fourth gusset members **14**A, **14**D. The vertical pile **21** has one pile surface **23-1** contacting the wall **13** of the collar portion **11** of the fixture **10**, and the vertical pile has another, opposing pile surface **23-2** contacting the medial surface **41** of the U-bolt **40**. It is to be appreciated that multiple U-bolts **40** may be provided and offset from one another vertically to cooperate with multiple pairs of holes defined in the first and fourth gusset members **14**A, **14**D.

FIG. **4**A is a top plan view of first and second fixtures **60-1**, **60-2** that define rectangular recesses **52** in collar portions **51** thereof, and that are joined together with fasteners **25** to form a pile connection apparatus **69** according to one embodiment in which a vertical pile **61** having a rectangular cross-sectional shape (e.g., with a central web **62** and two peripheral flanges **63** arranged perpendicular to the web **62**) is supported. Each fixture **60-1**, **60-2** includes a collar portion **51** with perpendicular wall extensions **51'** that bound a rectangular recess **52** (appearing rectangular when viewed from above), three micropile sleeves **18**A-**18**C, and multiple gusset members **54**A-**54**D projecting laterally outward (i.e., toward the micropile sleeves **18**A-**18**C). Connecting members **56**A-**56**C that are curved in shape are coupled between adjacent gusset members **54**A-**54**D, with the connecting members **56**A-**56**C being laterally spaced apart from the collar portion **51** to provide gaps **57** between the collar portion **51** and the connecting members **56**A-**56**C, with such gaps **57** each having a modified wedge shape and also extending between adjacent gusset members **54**A-**54**C. Each micropile sleeve **18**A-**18**C is directly coupled to a corresponding gusset member **54**A-**54**C, and each connecting member **56**A-**56**C is directly coupled to an adjacent pair of gusset members **54**A-**54**C. As shown in FIG. **4**A, the fasteners **25** extend through aligned holes (not shown) defined in the first and fourth gusset members **54**A, **54**D of the first and second fixtures **60-1**, **60-2**. Additionally, the vertical pile **61** has two laterally opposing pile surfaces **53-1**, **53-2** that are contacted by walls of the collar portions **51** of the fixtures **60-1**, **60-2**, respectively. In certain embodiments, each collar portion **51** surrounds 50% of a perimeter or projected top border area of the pile **61**, such that both collar portions **51** in combination surround an entire perimeter or projected top border area of the pile **61**. In certain embodiments, each collar portion surrounds a majority (e.g., in a range of 40 to 50 percent) of a perimeter or projected top border area of the vertical pile **61**. FIGS. **4**B-**4**D provide additional (i.e., front elevational, side elevational, and perspective) views of the pile connection apparatus **69** affixed to the pile **61**. The micropile sleeves **18**A-**18**C are each oriented with central axes non-parallel to a central axis of the vertical pile **61**. As shown, the gusset members **54**A-**54**D each have a generally triangular shape, with an increased height proximate to the collar portions **51** and a reduced height proximate to the micropile sleeves **18**A-**18**C. FIG. **4**D shows a lower portion of the vertical pile **61** extending below the collar portions **51** as being visible through the gaps **57** between the connecting members **56**A-**56**C and the gusset members **54**A-**54**C, but it is to be appreciated that in use, the connecting members **56**A-**56**C may be arranged at a ground surface, with a lower portion of the vertical pile **61** being driven downward into the ground and therefore obscured from view.

Although various preceding figures have shown vertical piles having generally circular and generally rectangular cross-sectional shapes or borders, it is to be appreciated that collar portions of fixtures and coupling members disclosed herein may be shaped to receive vertical piles of any suitable external cross-sectional shape.

In certain embodiments, micropile sleeves may be indirectly coupled to gusset members, such as by positioning one or more (inter-gusset) connecting members therebetween. Additionally, in certain embodiments, fasteners may extend through holes defined in outwardly projecting tab extensions of a collar portion and defined in inwardly projecting tab extensions of a connecting member.

FIG. **5**A is a top plan view of first and second fixtures **80-1**, **80-2** configured to be coupled together with fasteners **25** to form a pile connection apparatus **79** according to one embodiment. Each fixture **80-1**, **80-2** includes two gusset members **84**A, **84**B, a single connecting member **86** connecting (and extending past) the gusset members **84**A, **84**B, a collar portion **70**, and two micropile sleeves **78**A, **78**B (and optional mounting plates **77**A, **77**B for the micropile sleeves **78**A, **78**B) indirectly coupled to the gusset members **84**A, **84**B through the connecting member **86**. Each fixture **80-1**, **80-2** further includes fastener receiving holes defined in outwardly projecting tab extensions **74**A, **74**B of the collar portion **70** and defined in inwardly projecting tab extensions **85**A, **85**B of the connecting member **86**. Such a configuration does not require any holes to be defined in the gusset members **84**A, **84**B. Each collar portion **70** further includes a wall **73** bounding a recess **72** that is configured to receive and contact an outer surface portion of a vertical pile (not shown). For each fixture **80-1**, **80-2**, a gaps **87** is provided between the connecting member **86**, the collar portion **70**, and the gusset members **84**A-**84**B. FIG. **5**B shows a gusset member **84**A of each fixture **80-1**, **80-2**, showing that the gusset member **84**A has a length L, and has a proximal height (Hp) (at an edge adjacent to the collar portion) that is greater than the distal height HD (at an edge distal from the collar portion, and proximate to the connecting member **86**). Additionally, for each gusset member **84**A, **84**B, the distal height Hp is greater than the width W thereof.

FIG. **6**A is a top plan view of a fixture **90** joined by fasteners **25** to a coupling member **101** to form a pile connection apparatus **99** according to one embodiment. The fixture **90** includes a collar portion **91**, three gusset members **94**A-**94**C, a single connecting member **96** connecting the gusset members **94**A-**94**C, three micropile sleeves **78**A-**78**C (and optional mounting plates **77**A-**77**C for the micropile sleeves **78**A-**78**C) indirectly coupled to the gusset members **94**A-**94**C through the connecting member **96**. Fastener receiving holes are defined in two gusset members **94**A, **94**C and in projecting tab extensions **104**A, **104**B of the coupling member **101**. The coupling member **101** includes a wall **100** defining a recess **102** and the projecting tab extensions **104**A. **104**B thereof, with fastener receiving holes **106** (shown in FIG. **6**C) defined in the tab extensions of the coupling member **101**. FIG. **6**B shows a gusset member **94**A of the fixture **90**, showing fastener receiving holes **107**, **108**, as well as depicting a length L, a proximal height (Hp) (at an edge adjacent to the collar portion **91**) that is greater than the distal height HD (at an edge distal from the collar portion **91**, and proximate to the connecting member **96**). Additionally, for each gusset member **94**A-**94**C, the distal height Hp is greater than a width thereof. FIG. **6**C provides a perspective view of the coupling member **101** as described above.

FIG. **7**A is a top plan view of first and second fixtures **110-1**, **110-2** configured to be coupled together with fasteners (e.g., **25** as shown in FIG. **5**A) that are configured to extend through holes (defined in gusset members **14**A, **14**D at regions **126**) to form a pile connection apparatus according to one embodiment. Dashed lines **126**A represent paths for insertion of fasteners (e.g., **25** as shown in FIG. **5**A) to enable coupling between the fixtures **110-1**, **110-2**. Each fixture **110-1**, **110-2** includes five gusset members **114**A-**114**E, a single connecting member **116** connecting the gusset members **114**A-**114**E, a collar portion **111**, and four micropile sleeves **78**A-**78**D indirectly coupled to the gusset members **114**A-**114**E through the connecting member **116**. Fastener receiving holes (not shown) are defined in selected gusset members **114**A, **114**E. Each collar portion **111** has a wall **113** defining a recess **112** for receiving a portion of a vertical pile (not shown). FIG. **7**B shows a gusset member **114**A of the fixture **110**, showing fastener receiving holes **107**, **108**, as well as depicting a length L, a proximal height (Hp) (at an edge adjacent to the collar portion **111**) that is greater than the distal height HD (at an edge distal from the collar portion **111**, and proximate to the connecting member **116**). In use of the fixtures **110-1**, **110-2** of FIG. **7**A, a pile may be positioned vertically relative to a ground (and optionally, driven partially into the ground) to form a vertical pile; the fixtures **110-1**, **110-2** may be coupled (using fasteners extending through the holes **107**, **108**) to the vertical pile; and multiple micropiles may be inserted through the micropile sleeves **78**A-**78**D into the ground.

FIGS. **8**A-**8**D illustrate a first alternative pile connection apparatus **129** including two fixtures **130-1**, **130-2** affixed to a cylindrical vertical pile **21**, in which such figures provide side elevational, front elevational, top plan, and lower perspective views, respectively. The pile connection apparatus **129** includes two complementary fixtures **130-1**, **130-2** joined with fasteners **25**, with each fixture **130-1**, **130-2** having a collar portion **120** with a curved wall section **121**, and one gusset member **124** extending from the collar portion **120** to a micropile sleeve **128**. Each collar portion **120** includes outwardly projecting tab extensions **123** through which fastener receiving holes are defined. Unlike the embodiments shown in the preceding figures, the apparatus **129** lacks any connecting members spanning between micropile sleeves **128**. As shown, each gusset member **124** lies along an imaginary plane that is non-vertical, but that is parallel to a central axis A_2 (shown in FIG. **8**B) of the joined micropile sleeve **128**. The central axis A_2 of each micropile sleeve **128** is non-parallel to a central axis A_1 of the vertical pile **21**. To permit each gusset member **124** to be joined to the curved wall section **121** of the collar portion **120**, a proximal surface (or proximal edge) **125** of the gusset member **124** is curved in shape to substantially conform to a curvature of the curved wall section **121**.

FIGS. **9**A-**9**D illustrate a pile connection apparatus including a single fixture **140** with a collar **141** fitting around an entire perimeter of a vertical pile **21**, and including fasteners **25**A extending through the collar **141** and the vertical pile **21**. Two opposing gusset members **144**A-**144**B extend from the collar **141** to two micropile sleeves **148**A, **148**B (each configured to receive a corresponding micropile **151**A, **151**B as shown in FIG. **9**C), with each gusset member **144**A, **144**B lying along a respective imaginary plane that is inclined away from vertical. Each gusset member **144**A, **144**B has a curved proximal surface (or proximal edge) **145** to conform to a curved surface of the collar **141**.

It is to be appreciated that in any embodiments shown herein, gusset members may be arranged along imaginary planes (e.g., definable through an upper surface, a lower surface, a proximal surface adjacent to a collar portion, and a distal surface distal from a collar portion) that are non-vertical and that are substantially parallel to a sleeve axis of a micropile sleeve affixed to the gusset member. If such a gusset member is coupled to a collar portion having a curved wall, then a proximal surface (or proximal edge) of the gusset member may be curved to substantially conform to a curvature of the curved wall of the collar portion.

The present disclosure includes a method for supporting a vertical pile. The method comprises: positioning a pile vertically to a ground to form a vertical pile; coupling a pile connection apparatus as disclosed herein to the vertical pile, with the first collar portion of the first fixture receiving a first portion of the vertical pile, with the second collar portion of the second fixture receiving a second portion of the vertical pile, and with the plurality of fasteners cooperating with the first and second collar portions to cause the vertical pile to be frictionally engaged therebetween; inserting a plurality of first micropiles through the plurality of first micropile sleeves and into the ground; and inserting a plurality of second micropiles through the plurality of second micropile sleeves and into the ground. In certain embodiments, a pile connection apparatus may be coupled to the vertical pile after a portion of the vertical pile is driven into the ground. In certain embodiments, a pile connection may be coupled to the vertical pile before a portion of the vertical pile is driven into the ground.

Many modifications and other embodiments of the embodiments set forth herein will come to mind to one skilled in the art to which the embodiments pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the description and claims are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. It is intended that the embodiments cover the modifications and variations of the embodiments provided they come within the scope of the appended claims and their equivalents. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

What is claimed is:

1. A pile connection apparatus for supporting a vertical pile, the apparatus comprising:

a first fixture comprising a first collar portion, a plurality of first micropile sleeves, a plurality of first gusset members coupling the plurality of first micropile sleeves to the first collar portion, and at least one first connecting member coupling adjacent first gusset members of the plurality of first gusset members;

wherein:

the first collar portion comprises a first recess configured to receive a first portion of the vertical pile, the first recess being bounded by a first wall configured to contact a first pile surface of the first portion of the vertical pile;

the plurality of first gusset members project laterally outward from the first collar portion;

an entirety of the at least one first connecting member is arranged proximate to a perimeter of the first fixture and is laterally displaced relative to the first collar portion, with one or more gaps between the first collar portion and the at least one first connecting member;

the at least one first connecting member has a cross-sectional height extending continuously in a vertical direction, and has a cross-sectional width extending continuously in a lateral direction oriented perpendicular to the vertical direction and oriented radially outward relative to the first collar portion, wherein the cross sectional height is greater than the cross-sectional width; and

one or more of (i) the first collar portion and (ii) at least one first gusset member of the plurality of first gusset members comprises and/or is configured to receive at least one fastener to cause the first wall to engage the first pile surface.

2. The pile connection apparatus of claim **1**, wherein the plurality of first gusset members are intermediately coupled to the plurality of first micropile sleeves, with a first connecting member of the at least one first connecting member arranged between adjacent first gusset members of the plurality of first gusset members.

3. The pile connection apparatus of claim **1**, wherein:

each first gusset member of the plurality of first gusset members comprises: (i) a length extending in a direction from the first collar portion to a corresponding first micropile sleeve of the plurality of first micropile sleeves, (ii) a width orthogonal to the length, (iii) a proximal height at a point of contact with the first collar portion, and (iv) a distal height at a point of contact with the at least one first connecting member; and

for each first gusset member, the proximal height is greater than the distal height, and the distal height is greater than the width.

4. The pile connection apparatus of claim **1**, wherein:

each first micropile sleeve of the plurality of first micropile sleeves has a first sleeve inlet, a first sleeve outlet, and a first sleeve axis extending through the first sleeve inlet and the first sleeve outlet; and

for each first micropile sleeve, the first sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus.

5. The pile connection apparatus of claim **4**, wherein:

each first gusset member of the plurality of first gusset members comprises: a first upper surface, a first lower surface, a first proximal surface proximal to the first collar portion, and a first distal surface distal from the first collar portion;

for each first gusset member, a first imaginary plane is definable through the first upper surface, the first lower surface, the first proximal surface, and the first distal surface; and

for each first gusset member, the first imaginary plane is non-vertical, and is substantially parallel to the first sleeve axis of the first micropile sleeve coupled to the first gusset member.

6. The pile connection apparatus of claim **5**, wherein:

the first wall is curved in shape, with the pile connection apparatus being configured to support a vertical pile having a cylindrical shape; and

the first proximal surface of each first gusset member abuts the first wall, with the first proximal surface being curved in shape to substantially conform to a curvature of the first wall.

7. The pile connection apparatus of claim **1**, further comprising:

a second fixture comprising a second collar portion, a plurality of second micropile sleeves, a plurality of second gusset members coupling the plurality of second micropile sleeves to the second collar portion, and at least one second connecting member coupling adjacent second gusset members of the plurality of second gusset members;

wherein:

the second collar portion comprises a second recess configured to receive a second portion of the vertical pile, the second recess being bounded by a second wall configured to contact a second pile surface of the second portion of vertical pile, wherein the second portion of the vertical pile opposes the first portion of the vertical pile;

the plurality of second gusset members project laterally outward from the second collar portion;

an entirety of the at least one second connecting member is laterally displaced relative to the second collar portion;

one or more of (iii) the second collar portion and (iv) the at least one second gusset member of the plurality of second gusset members comprises and/or is configured to receive the at least one fastener to cause the second wall to engage the second pile surface, with the second recess of the second fixture opposing the first recess of the first fixture.

8. The pile connection apparatus of claim **7**, wherein:

each second micropile sleeve of the plurality of second micropile sleeves has a second sleeve inlet, a second sleeve outlet, and a second sleeve axis extending through the second sleeve inlet and the second sleeve outlet; and

for each second micropile sleeve, the second sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus.

9. The pile connection apparatus of claim **7**, wherein:

the first recess is configured to receive the first portion of the vertical pile extending around a first perimetral portion spanning a range of 40 to 50 percent of a perimeter of the vertical pile, and the second recess is configured to receive the second portion of the vertical pile extending around a second perimetral portion spanning a range of 40 to 50 percent of the perimeter of the vertical pile.

10. The pile connection apparatus of claim **7**, wherein;

a plurality of first fastener openings are defined in one or more of (i) the first collar portion and (ii) the at least one first gusset member of the plurality of first gusset members, and a plurality of second fastener openings is defined in one or more of (iii) the second collar portion and (iv) the at least one second gusset member of the plurality of second gusset members,

the at least one fastener comprises a plurality of fasteners, and

each first fastener opening is configured to be aligned with a corresponding second fastener opening to receive one fastener of the plurality of fasteners extending through the aligned first and second fastener openings.

11. The pile connection apparatus of claim **7**, wherein:

one or more fasteners of the at least one fastener is affixed to or integrated with one or more of (i) the first collar portion, (ii) at least one first gusset member of the plurality of first gusset members, (iii) the second collar portion, and (iv) the at least one second gusset member of the plurality of second gusset members.

12. The pile connection apparatus of claim 1, wherein the plurality of first gusset members comprises at least three first gusset members.

13. The pile connection apparatus of claim 1, wherein one or more of (i) the first collar portion and (ii) the at least one first gusset member of the plurality of first gusset members comprises fastener openings configured to receive at least one U-bolt that is configured to extend around a portion of a perimeter of the vertical pile.

14. A pile connection apparatus for supporting a vertical pile, the apparatus comprising:

a first fixture comprising a first collar portion, a plurality of first micropile sleeves, a plurality of first gusset members coupling the plurality of first micropile sleeves to the first collar portion, and at least one first connecting member coupling adjacent first gusset members of the plurality of first gusset members,

wherein:

the first collar portion comprises a first recess configured to receive a first portion of the vertical pile, the first recess being bounded by a first wall configured to contact a first pile surface of the first portion of the vertical pile;

the plurality of first gusset members project laterally outward from the first collar portion;

an entirety of the at least one first connecting member is arranged proximate to a perimeter of the first fixture and is laterally displaced relative to the first collar portion, with one or more gaps between the first collar portion and the at least one first connecting member; and

the at least one first connecting member has a cross-sectional height in a vertical direction, and has a cross-sectional width in a lateral direction extending perpendicular to the vertical direction and extending radially outward from the first collar portion, wherein the cross sectional height of the at least one first connecting member is greater than the cross-sectional width thereof;

a second fixture comprising a second collar portion, a plurality of second micropile sleeves, a plurality of second gusset members coupling the plurality of second micropile sleeves to the second collar portion, and at least one second connecting member coupling adjacent second gusset members of the plurality of second gusset members;

wherein:

the second collar portion comprises a second recess configured to receive a second portion of the vertical pile, the second recess being bounded by a second wall configured to contact a second pile surface of the second portion of the vertical pile, wherein the second portion of the vertical pile opposes the first portion of the vertical pile;

the plurality of second gusset members project laterally outward from the second collar portion;

an entirety of the at least one second connecting member is arranged proximate to a perimeter of the second fixture and is laterally displaced relative to the second collar portion, with one or more gaps between the second collar portion and the at least one second connecting member; and

the at least one second connecting member has a cross-sectional height in a vertical direction, and has a cross-sectional width in a lateral direction extending perpendicular to the vertical direction and extending radially outward from the second collar portion, wherein the cross sectional height of the at least one second connecting member is greater than the cross-sectional width thereof; and

a plurality of fasteners configured to cooperate with the first fixture and the second fixture to cause the first wall to engage the first pile surface, to cause the second wall to engage the second pile surface, and to cause the vertical pile to be frictionally engaged between the first fixture and the second fixture.

15. The pile connection apparatus of claim 14, wherein:

each first gusset member of the plurality of first gusset members comprises: (i) a first length extending in a direction from the first collar portion to a corresponding first micropile sleeve of the plurality of first micropile sleeves, (ii) a first width orthogonal to the length, (iii) a first proximal height at a point of contact with the first collar portion, and (iv) a first distal height at a point of contact with the at least one first connecting member;

each second gusset member of the plurality of second gusset members comprises: (i) a second length extending in a direction from the second collar portion to a corresponding second micropile sleeve of the plurality of second micropile sleeves, (ii) a second width orthogonal to the length, (iii) a second proximal height at a point of contact with the second collar portion, and (iv) a second distal height at a point of contact with the at least one second connecting member;

for each first gusset member, the first proximal height is greater than the first distal height, and the first distal height is greater than the first width; and

for each second gusset member, the second proximal height is greater than the second distal height, and the second distal height is greater than the second width.

16. The pile connection apparatus of claim 14, wherein:

each first micropile sleeve of the plurality of first micropile sleeves has a first sleeve inlet, a first sleeve outlet, and a first sleeve axis extending through the first sleeve inlet and the first sleeve outlet;

each second micropile sleeve of the plurality of second micropile sleeves has a second sleeve inlet, a second sleeve outlet, and a second sleeve axis extending through the second sleeve inlet and the second sleeve outlet;

for each first micropile sleeve, the first sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus; and

for each second micropile sleeve, the second sleeve axis is non-parallel to the vertical pile to be supported by the pile connection apparatus.

17. The pile connection apparatus of claim 16, wherein:

each first gusset member of the plurality of first gusset members comprises: a first upper surface, a first lower surface, a first proximal surface proximal to the first collar portion, and a first distal surface distal from the first collar portion;

for each first gusset member, a first imaginary plane is definable through the first upper surface, the first lower surface, the first proximal surface, and the first distal surface;

for each first gusset member, the first imaginary plane is non-vertical, and is substantially parallel to the first sleeve axis of the first micropile sleeve coupled to the first gusset member;

each second gusset member of the plurality of second gusset members comprises: a second upper surface, a second lower surface, a second proximal surface proximal to the second collar portion, and a second distal surface distal from the second collar portion;

for each second gusset member, a second imaginary plane is definable through the second upper surface, the second lower surface, the second proximal surface, and the second distal surface; and

for each second gusset member, the second imaginary plane is non-vertical, and is substantially parallel to the second sleeve axis of the second micropile sleeve coupled to the second gusset member.

18. The pile connection apparatus of claim **17**, wherein:

the first wall is curved in shape, with the pile connection apparatus being configured to support the vertical pile having a cylindrical shape;

the first proximal surface of each first gusset member abuts the first wall, with the first proximal surface being curved in shape to substantially conform to a curvature of the first wall;

the second wall is curved in shape, with the pile connection apparatus being configured to support the vertical pile having a cylindrical shape; and

the second proximal surface of each second gusset member abuts the second wall, with the second proximal surface being curved in shape to substantially conform to a curvature of the second wall.

19. The pile connection apparatus of claim **14**, wherein:

the vertical pile configured to be supported by the pile connection apparatus comprises a pile cross-sectional width or diameter; the first recess comprises a maximum horizontal depth with a dimension in a range of 40 to 50 percent of the pile width or diameter; and the second recess comprises a maximum horizontal depth with a dimension in a range of 40 to 50 percent of the pile width or diameter.

20. The pile connection apparatus of claim **19**, wherein:

a plurality of first fastener openings are defined in one or more of (i) the first collar portion and (ii) at least one first gusset member of the plurality of first gusset members, a plurality of second fastener openings is defined in one or more of (iii) the second collar portion and (iv) at least one second gusset member of the plurality of second gusset members, and

each first fastener opening is configured to be aligned with a corresponding second fastener opening to receive one fastener of the plurality of fasteners extending through the aligned first and second fastener openings.

21. The pile connection apparatus of claim **14**, wherein one or more fasteners of the plurality of fasteners is affixed to or integrated with one or more of (i) the first collar portion, (ii) at least one first gusset member of the plurality of first gusset members, (iii) the second collar portion, and (iv) at least one second gusset member of the plurality of second gusset members.

22. A method for supporting a vertical pile, the method comprising:

positioning a pile vertically to a ground to form a vertical pile;

coupling a pile connection apparatus according to claim **14** to the vertical pile, with the first collar portion of the first fixture receiving a first portion of the vertical pile, with the second collar portion of the second fixture receiving a second portion of the vertical pile, and with the plurality of fasteners cooperating with the first and second collar portions to cause the vertical pile to be frictionally engaged therebetween;

inserting a plurality of first micropiles through the plurality of first micropile sleeves and into the ground; and

inserting a plurality of second micropiles through the plurality of second micropile sleeves and into the ground.

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